BREAKING UP IS HARD TO DO: WHY FIRMS FRAGMENT PRODUCTION ACROSS LOCATIONS

by

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Abstract

This paper documents the relative importance of labor cost differences, distance to suppliers, and communication technology in a firm's domestic and foreign sourcing decisions. Using an original dataset of U.S. manufacturers' decisions to contract for manufacturing services, I show that domestic fragmentation: i) is far more prevalent than offshoring; ii) changes firms' opportunity cost to offshore; and iii) is facilitated by communication technology. In contrast, communication technology does not necessarily lead firms to offshore. Firms fragment production to access cheaper labor, and countries that offer significant labor cost savings tend not to have the technology infrastructure to support high-tech production.

Keyword*:* fragmentation, offshoring, technology, contract manufacturing services

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International production fragmentation, or offshoring, has significant domestic employment, wage, and productivity ramifications (e.g., Feenstra and Hanson, 1999; Jones and Kierzkowski, 2001; Grossman and Rossi-Hansberg, 2008; Rodríguez-Clare, 2010). Despite great interest in offshoring and multinational sourcing, little is known about how and why firms break up their production process across space. There are two dimensions to firms' sourcing strategies: (a) whether to fragment production; and (b) conditional on fragmentation, whether to source domestically or offshore. These decisions are shaped by production costs (e.g., relative wages), transportation costs, and coordination/communication costs. While advances in communication technology are often credited with increased offshoring, the lack of firm-level fragmentation data has made this relationship difficult to assess empirically. There is even less evidence on domestic sourcing, though it may be an alternative to offshoring with very different implications for national welfare.

This paper exploits a novel micro-level dataset to examine how labor costs, transportation costs, and communication technology explain firm-level fragmentation decisions. The new data are from the U.S. Census Bureau's 2007 Census of Manufactures. They identify whether a plant purchased contract manufacturing services from other plants (within its company or from another company) to process its inputs; and if so, whether the plant primarily purchased these services domestically or abroad. Contract manufacturing services (CMS) cover the fragmentation of inputs that are customized to meet a firm's specific production criteria. Although CMS purchases are only a subset of all possible fragmentation, the fact that they require communicating production specifications across locations makes them uniquely suited to assess the role of communication technology in fragmentation. Most importantly, these data provide a direct measure of fragmentation that covers plants' domestic and foreign fragmentation decisions.

I combine the CMS data with additional information from the Census of Manufactures, the Longitudinal Business Database, and U.S. Customs import transactions, to document seven new facts about plants that fragment their customized production process. The facts reveal significant variation in firms' sourcing strategies. A substantial share of plants do not purchase CMS, and of those that do, the majority primarily source from domestic suppliers. The share of plants that fragments domestically is 13 times higher than the share of plants that offshores, and domestic fragmenters employ almost nine times more workers. The data also show that fragmenting plants are larger and more productive, particularly so if they offshore.[1]

[1]The finding of a productivity premia for offshorers is consistent with Kurz (2006) who analyzes U.S. plant-level data on foreign purchases of materials by manufacturers in 1987 and 1992; and

These aggregate patterns hold within industries so they cannot be not explained by sectoral differences in production requirements. There is also significant geographical variation in the distribution of fragmenting plants that is consistent with a role for local labor costs in fragmentation decisions. 30 percent of plants in high-wage states fragment domestically, compared to only 22 percent in low-wage states. Finally, offshoring firms' share of imports from low-income countries is about twice as large as low-income import shares for non-offshorers.

I incorporate these findings into a model of heterogeneous firms that can fragment production across domestic and foreign locations. Firms fragment production to access cheaper labor, but doing so involves a fixed cost (as in Antràs and Helpman, 2004) as well as per-task transportation and coordination costs (as in Grossman and Rossi-Hansberg, 2008). The model delivers standard predictions about fixed costs and productivity sorting. In addition, production cost savings (and therefore profitability) from fragmentation are increasing in a plant's local wage and decreasing in distance to suppliers. The key feature of the model is that the cost function also provides a simple framework in which firm use of communication technology lowers fragmentation costs, but its impact depends on the extent to which the firm's (industry-level) production process can be digitized into an electronic format. The cost function also allows for the possibility that the effectiveness of a firm's technology depends on its suppliers' technological capabilities.

To assess the model's predictions, I jointly estimate the probability that a plant will fragment production domestically or offshore. I employ a nested logit model in which the upper level decision is whether to fragment or not, and the lower level decision is whether to fragment domestically or offshore. This nesting structure does not imply a sequential order to firms' decisions, but does allow for unobserved correlations in the errors of plants that choose to fragment. The estimation technique yields four main results. First, higher local labor costs increase the probability of fragmentation. A plant in Washington state is almost four percentage points more likely to fragment than the "same" plant in Mississippi, where average production worker wages are about 30 percent lower. Second, plants that are further away from potential domestic suppliers or foreign entry points are less likely to fragment. Third, more productive plants are more likely to fragment production, and conditional on fragmentation, more likely to offshore. Finally, plant use of electronic networks to coordinate shipments (as a measure of communication technology) is associated with an increased probability of domestic (foreign) fragmentation of nine (one) percentage

Tomiura (2007) who creates productivity rankings for Japanese firms' sourcing choices. This paper extends those findings by showing they hold for fragmentation of customized inputs.

points. These results are all robust to controlling for variation in local skills, demand, manufacturing concentration, and endogenous location choice.

A plant's use of communication technology may depend upon the fragmentation strategy it plans to adopt. To address the potential for reverse causality, I assess whether the impact of electronic networks depends upon a firm's ability to digitize its input requirements in an electronic format. If electronic communication truly lowers fragmentation costs by facilitating communication about production requirements, then it will have a bigger impact on firms in industries in which those production requirements can be more readily digitized. Conversations with contract manufacturing suppliers revealed that communication technology lowers fragmentation costs more when it can be used in conjunction with Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) software.[2] I therefore measure industry variation in the feasibility of digitizing input requirements as the share of plants in an industry that used Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) in 1999. CAD/CAM enables plants to codify specifications in an electronic format that can be sent to suppliers, but the extent to which plants can use CAD/CAM depends upon their industry's production process characteristics. Since CAD/CAM is an industry-level variable that depends upon physical characteristics of the production process, it is largely exogenous at the plant level.

The estimates reveal a strong differential impact of electronic networks across industry CAD/CAM intensity. Plants using networks in the most CAD/CAM intensive industries are almost twice as likely to fragment production domestically as plants using networks in the least CAD/CAM intensive industries. This differential impact does not depend on firms' (endogenous) decision to use networks and thus provides evidence that communication technology facilitates domestic fragmentation by lowering coordination costs. In contrast, the estimates show that the probability that plants using electronic networks will offshore is *decreasing* in industry CAD/CAM intensity. A comparable analysis using a linear probability model reveals the same basic patterns. Relative to plants that do not use electronic networks, plants that use networks in "other apparel" manufacturing (CAD intensity of 4 percent) are only 4.5 percentage points more likely to fragment, while plants using networks in the "semiconductor machinery" manufacturing (CAD intensity of 98 percent) are over 30

[2]For example, a gasket producer described costs that fell from "hundreds of dollars to cents" when CAD files that could be plugged into CAM programs replaced conventional drawings, sent via fax, that required additional verbal communication to interpret. In contrast, a spring manufacturer whose production process does not work with CAM software did not experience these cost reductions from electronic communication. These conversations took place at Mid-Atlantic Design-2 Part Shows in April 2011 and November 2010.

points more likely to fragment than plants not using networks. However, as in the nested logit results, fragmenting plants that use networks in the most CAD/CAM intensive industries are *less* likely to source offshore. The model provides a potential explanation for this finding: the effectiveness of firms' technology depends upon suppliers' ability to receive and utilize electronic communications about the production process.

The fact that offshoring firms source disproportionately from low-income countries is consistent with a labor cost savings motive, but also suggests that prime offshore sourcing locations may not have the technological capabilities to support high-tech production processes. Consistent with this hypothesis, estimates using firm-country level import data show that firm use of communication technology and industry CAD/CAM intensity both increase the probability of sourcing from high-tech countries like Japan, but decrease the likelihood of sourcing from low-tech countries like Bangladesh. The evidence therefore supports the premise that communication technology facilitates coordination across locations, but it also shows that communication technology does not necessarily lead firms to offshore. Although surprising, this result is intuitive since higher technology countries tend not to have the lower wages which provide the motivation for firms to offshore in the first place.

This paper contributes to two distinct literatures. First, it extends the literature on vertical production networks. Existing evidence on vertical networks is often limited to sectoral imputations from input-output tables (e.g., Hummels, Ishii and Yi, 2001; Johnson and Noguera, 2012) or to examining the intensive margin of multinational firms' activities (e.g., Hanson, Mataloni and Slaughter, 2005; Yeaple, 2003). This paper uses a direct firm-level measure that captures fragmentation inside and outside the boundary of the firm. To my knowledge, it is the first to analyze firms' purchases of contract manufacturing services purchases.[3] In addition, it also provides the first direct evidence on the role of communication technology in firms' fragmentation decisions.[4]

Most importantly, this paper sheds new light on vertical networks both within and across national borders. Empirical work on domestic vertical networks is limited due to a lack of available data. Several papers exploit the U.S. Census Bureau Commodity

[3]See Kamal, Moulton and Ribarsky (2013) for a discussion of efforts to collect additional data of firms' contract manufacturing services purchases.

[4]Feinberg and Keane (2006) attribute the majority of increased intra-firm trade between the U.S. and Canada from 1983 to 1996 to changes in technology. In that paper technical change is simply the residual from a structural model. The analysis here uses a direct measure of communication technology and provides firm-level evidence on a specific channel through which this technology facilitates fragmentation.

Flow Survey data on shipments from U.S. plants to zip codes or export ports to analyze integrated domestic vertical networks (Atalay, Hortacsu and Syverson, 2013) and the role of distance on plants' shipments (Hillberry and Hummels, 2008; Holmes and Stevens, 2012).[5] Holmes and Stevens (2012) show that distance is an important factor not just for exports, but also for domestic shipments–a result that highlights the importance of analyzing exports relative to the same type of domestic activity. In this paper, I analyze the role of distance, labor costs, and communication technology on both domestic and foreign sourcing decisions. To my knowledge, this is the first analysis that covers firms' domestic and foreign sourcing decisions for the same type of fragmentation.

This paper also contributes to the literature on offshoring and technology. A number of theoretical models either predict or assume a strong role for communication technology in offshoring decisions (e.g., Antràs, Garicano and Rossi-Hansberg, 2006; Grossman and Rossi-Hansberg, 2008; Baldwin and Venables, 2010), but there is almost no evidence documenting this relationship. This paper finds that electronic communication is an important factor in firms' extensive margin decision to fragment production. However, the data also show that the impact of this technology varies systematically across industries, having a much bigger impact in IT-intensive industries. In addition, the type of communication technology studied here increases the likelihood that firms in the most IT-intensive industries will fragment domestically, but decreases the likelihood that they will offshore.

In the next section, I describe the data and present seven new stylized facts about plants and firms that fragment their production. In Section 2, I incorporate these facts into a theoretical framework in which firms fragment production to access cheaper labor, but face transportation and coordination costs from doing so. In Section 3, I use the new data to estimate the role of wage differences, distance to supplier, and communication technology in firms' fragmentation and offshoring decisions, and to assess the role of country-level technology in firms' sourcing decisions. Section 4 shows that variation in communication technology accounts for more of the explained variation in fragmentation and offshoring decisions than wage differences and distance combined. Section 5 concludes.

[5]The Commodity Flow Survey data provide information on a random sample of shipments from a sample of manufacturing, wholesale, and mining establishments over the course of a particular week in each quarter of the year.

1 New data and facts on fragmentation

One contribution of this paper is to use the confidential plant-level U.S. Census Bureau data to provide a set of new facts about manufacturing plants that fragment their production process across locations. The fragmentation data are based on a question asked in the 2007 Census of Manufactures (CM). The CM is conducted in years that end in 2 and 7. It covers the universe of manufacturing establishments, though the fragmentation data are available for a subset of plants that covers 75 percent of manufacturing sales. Details of the CMS sample and corrections for possible sample selection are in the data appendix.

The new question asks: "Did this establishment purchase contract manufacturing services from other companies or other establishments or your company to process materials or components that this establishment owns or controls?" Establishments that answer yes are also asked whether they primarily purchase these services domestically or abroad.[6] Contract manufacturing services (CMS) is a term used by practitioners that covers input production or assembly that is customized by the supplier to meet specifications provided by the fragmenting plant. The data in the paper therefore cover customized fragmentation of production, but do not include fragmentation of standardized, off-the-shelf inputs. Since the former necessarily requires communicating specifications across locations, the data are uniquely suited to assess the role of communication technology in fragmentation.

Although the CMS data only cover customized production processes, they provide new information about fragmentation along several key dimensions. First, the data cover both domestic and foreign fragmentation for the same measure. Second, they are collected at the plant level and reveal substantial intra-industry heterogeneity in plants' and firms' fragmentation decisions. Third, the data contain both outsourced and integrated production, unlike measures of foreign direct investment that consist only of the latter.[7] Fourth, the data do not rely on imported intermediates as a measure of foreign fragmentation. Intermediate trade measures often rely on input-output tables that are relatively coarse and therefore unlikely to identify intermediate inputs exclusively or entirely. In addition, imported intermediates are not a complete

[6] An establishment denotes a single physical location where business transactions take place and for which payroll and employment records are kept. It is synonymous with a plant. The question as it appeared on the Census form is presented in the data appendix.

[7] The inclusion of both types of fragmentation is important since, as noted in Feenstra (1998), "looking within multinational firms alone does not give a full perspective on what is happening" (p. 36). A limitation of the data is that they do not distinguish between fragmentation that takes place within or outside the firm so that it is not possible to investigate optimal firm boundaries.

offshoring measure since they exclude any final goods that are assembled overseas, while they may include imports of commodities or standardized inputs that are not part of a manufacturer's production process. The CMS data contain none of this ambiguity and clearly reflect offshoring activity.

The CM also provides information on plants' manufacturing sales, value-added, employees and industry. I match the CM data to the Business Register to identify plants' latitude and longitude. Using a firm identifier, I also aggregate the data to the firm level.[8] While the question design means that a single plant can only source primarily domestically or primarily offshore, multi-unit firms can potentially do both. Firms with at least one plant that purchases domestic CMS and at least one plant that purchases foreign CMS are therefore classified as "Domestic and Offshore Purchases." Finally, I link the firm-level data to U.S. Customs import transactions to identify the values and source countries of firms' imports.[9]

1.1 Economic significance of customized fragmentation

The CMS question does not ask plants about the magnitude of their CMS purchases. I therefore use two related measures to provide insight into the economic significance of the customized fragmentation studied in this paper. First, the CM asks plants to report "the cost of work done for you by others on your materials." These costs represent eight percent of input costs for plants that primarily offshore and 11 percent of costs for plants that fragment domestically.[10] Since these numbers exclude manufacturing services performed on inputs that the fragmenting plant does not own but instead simply controls or specifies, they suggest that CMS purchases constitute a non-trivial amount of plants' costs. At the firm level, imports also provide a metric to assess the importance of CMS purchases. The value of offshoring firms' imports constitute 20 percent of their sales, while firms that fragment primarily domestically have imports equal to just three percent of sales.[11] Finally, fragmenting firms' imports account for 77 percent of all manufacturing imports by manufacturing firms. Although these imports include final goods or other non-CMS products, the large

[8]The census data have an "alpha" variable that identifies the firm to which a given establishment belongs. This variable is superior to the employer identification number (EIN) used in other datasets to identify ownership. Since a single firm can use multiple EINs to file its tax returns, EINs may only identify part of a large firm.

[9]It is not possible to link the trade transactions data to individual establishments for multi-unit firms. See Bernard, Jensen and Schott (2009) for a detailed description of the import data.

[10]Input costs consist of costs for contract work, materials and parts, containers and packaging.

[11]I cannot measure input costs at the firm level, so I use sales instead.

fraction highlights the relevance of CMS purchasers in aggregate trade statistics.

1.2 New facts about fragmentation and offshoring

Most plants do not fragment customized production processes. The first column of Table 1 shows that 27 percent of plants fragment primarily domestically, while only two percent fragment primarily offshore.[12] Columns 2 and 3 provide the share of sales and employment respectively by plants' CMS purchase status. The percentages of sales and employment doubles to four percent for offshoring plants and jumps to 39 percent and 35 percent respectively for plants that fragment domestically. These findings are summarized as the following fact:

Fact 1: The majority of plants do not fragment their production process for customized inputs, even on a sales or employment-weighted basis.

To assess whether fragmentation heterogeneity is driven by industry characteristics, Table 3 presents the industry distribution of the shares of plants that purchase CMS domestically and offshore. I calculate participation shares within each of the 86 four-digit North American Classification System (NAICS) manufacturing industries. The first column shows that there are two industries in which no plants offshore production. In one of these industries, 10-20 percent of the plants purchase CMS domestically, while in the other industry 20-35 percent of plants do. One striking observation from Table 3 is that all industries have a positive share of plants that purchase domestic CMS. At least five percent of plants in every industry fragment domestically. Table 3 also shows substantial non-participation in every industry. The highest observed share of fragmenting plants is almost 60 percent.[13] Examining the diagonal of Table 3, it is evident that every industry has a higher share of domestic fragmenters than offshoring plants. This leads to:

Fact 2: Domestic fragmentation is more prevalent than offshoring, both across and within industries.

Almost half of the plants in the CMS sample belong to multi-unit firms. The right

[12] Limited offshore sourcing for customized production is in line with results in Tomiura (2007), who finds that only 2.68 percent of Japanese manufacturing firms outsource production offshore. There is almost no existing evidence on plants' domestic fragmentation, but Fally (2012) uses aggregate input-output tables to calculate the average number of sequential stages of domestic production.

[13] See the data appendix for summary statistics about plant participation shares within six-digit NAICS industries.

panel of Table 1 presents CMS participation shares at the firm level. While the fraction of fragmenting firms is very similar to the plant-level shares, the shares of firm-level employment and sales are significantly higher. Firms that purchase CMS account for 69 percent of manufacturing sales and 58 percent of employment. Despite this difference, *Fact 2* still holds at the firm level–domestic fragmentation is more prevalent than offshoring. Firms with one or more plants that purchase CMS primarily domestically account for 55 percent of manufacturing employment, while firms with at least some offshoring plants account for 19 percent. These activity-weighted firm participation shares suggest that production fragmentation is an important phenomenon in economic activity. While not all plants within a firm fragment production, fragmenting firms account for the majority of domestic manufacturing employment and sales. This finding is summarized by:

Fact 3: The majority of U.S. manufacturing sales and employment takes place at firms with at least one plant that purchases CMS.

Table 2 presents means for plant sales, employment and the log of value-added labor productivity by CMS purchase status. Columns 1 and 2 show that the average plant that fragments production is larger in terms of both sales and employment than the average non-fragmenting plant. In addition, plants that fragment production offshore are larger than domestic fragmenters. Average sales at a manufacturing plant with no CMS purchases are approximately $19 million, while average sales at an offshoring plant are over $50 million. Column 3 shows a similar pattern for productivity. Domestic purchasers are more productive than plants with no fragmentation, while offshorers are the most productive. To make meaningful comparisons of these variables across plants in different industries, I calculate a relative measure $x_{i,g}/\bar{x}_g$, where $\bar{x}_g$ is the mean of variable x for the six-digit NAICS industry g. Columns 4-6 show that the same patterns hold within industries. On average, plants that do not purchase CMS are smaller and less productive than their industry mean, while plants that purchase CMS domestically are above the industry mean; and offshorers are the largest and most productive plants. These results lead to two more stylized facts:

Fact 4: Plants that fragment production are larger and more productive than non-fragmenters.

Fact 5: Plants that fragment production offshore are larger and more productive than domestic fragmenters.

Plant participation shares also vary across U.S. states. I calculate the average produc-

tion worker wage per state and classify states as low wage, medium wage, and high wage depending upon the tercile to which their wage corresponds.[14] Two percent of plants offshore in all wage categories. In contrast, 22 percent of plants purchase CMS domestically in low wage states compared to 25 percent in middle wage states and 30 percent in high wage states. The finding leads to an additional stylized fact:

Fact 6: The share of plants in a state that fragment domestically is increasing in the state wage.

Although the CMS data lack specific details about the locations from which firms fragment, the trade data provide information about the countries from which firms import. Firms with no CMS purchases and domestic CMS purchases import 28 and 19 percent of their manufactured good imports from low-income countries respectively. In contrast, offshorers source 48 percent of their imports from low-income countries.[15] This leads to a final fact:

Fact 7: Offshoring firms import relatively more from low-income countries than domestic fragmenters and non-fragmenters.

The facts in this section highlight the importance of CMS purchasers in aggregate activity, but reveal significant heterogeneity in firms' sourcing decisions. In particular, firms' sourcing strategies cannot be fully explained by their industry characteristics and vary systematically with local wage conditions. In the next section, I incorporate these findings into a simple theoretical framework to guide the subsequent empirical analysis.

2 A model for domestic and foreign fragmentation

Let E denote aggregate expenditure in a representative industry (I omit industry subscripts for notational simplicity). Preferences across varieties for the representative industry have the standard CES form, with an elasticity of substitution $\varepsilon = \frac{1}{1-\sigma} > 1$.

[14] Details of the wage variable are in Section 3 and the online appendix. I use detailed production worker occupation data to construct industry-state level wages that are independent of compositional differences in the number of workers in occupations across states.

[15] See the online data appendix for a full analysis of the import data and how they link to the CMS data.

These preferences lead to demand for a particular variety i in a given industry,

$$q(i) = Ap(i)^{-\varepsilon},\ A = \frac{E}{\int_{i\in j} p(i)^{1-\varepsilon} di} \tag{2.1}$$

where $p(i)$ is the price of variety i and A is exogenous to an individual firm.

Labor is the only factor of production and is supplied inelastically. Producers use one unit of labor to produce one unit of task output. Production requires a continuum of tasks, indexed by k. Producers combine task output via a Leontief production function to produce a single composite input M, as in Rodríguez-Clare (2010). More formally, $M = min_k\{m_k\}$, $k \in [0, 1]$, where m_k denotes the output of task k.[16] For expositional simplicity, I normalize the number of tasks in the representative industry to one. Since tasks are defined by a unit labor requirement, the empirical analysis controls for the potential that industries differ in the number of tasks required to produce M. Producers have heterogeneous productivity, denoted by $\varphi > 0$, and transform the composite input M into their product via: $q = \varphi M$.

2.1 Profits with no fragmentation

With CES preferences, the optimal final good price is a mark-up over marginal cost given by $p_i(\varphi) = C_i/\varphi\sigma$, where C_i denotes the marginal cost of the input M for firm i. Let w_h denote the wage in the producer's home state. Because producers make m_k one-to-one from labor, the cost of one unit of M at the integrated producer is $C_i = w_h$ and its profits are:

$$\pi_I = \frac{(1-\sigma)A}{\sigma^{(1-\varepsilon)}} \left[\frac{\varphi}{w_h}\right]^{(\varepsilon-1)}. \tag{2.2}$$

2.2 Profits with fragmentation

Fragmentation allows producers to purchase task output from a manufacturing service provider (MSP) in another location with potentially lower labor costs.[17] The

[16] The assumption of no substitutability between tasks that use the same factor of production is common in the literature and simplifies the analysis. The model could be extended so that the composite input is produced via a constant elasticity of substitution technology that depends on the intensity with which each task is performed.

[17] MSP is the term used by practitioners and the U.S. Census Bureau to describe these suppliers.

assumption that wage differences exist within a country for the same quality of labor is supported by empirical evidence (e.g., Bernard, Redding and Schott, 2013). I assume perfect competition among MSPs so that the price of a task purchased from an MSP in another domestic (D) or offshore (O) sourcing location s is given by $P_s(m_k) = w_s$, where $s \in \{D, O\}$.[18]

While fragmentation allows a producer to access cheaper labor, it also entails certain costs. Establishing a supply network incurs a fixed cost f_D when the MSP is domestic and f_O when the MSP is foreign, with $f_D < f_O$. Fragmentation also incurs a task specific cost due to the additional transportation and coordination needs associated with breaking up the production function across locations. The fragmentation cost for firm i in industry j to source task k from location s is represented by the function:

$$\tau(\delta_{is}, \omega_k, \eta_i, \eta_s, \rho_j) \geq 1, \tag{2.3}$$

which I assume is continuously differentiable in all its arguments. δ_{is} denotes the distance between the final good producer and the sourcing location s. Transportation costs are increasing in distance so that $\frac{\partial \tau}{\partial \delta} > 0$. ω_k represents an inherent characteristic, such as weight or complexity, of the output from task k. $\frac{\partial \tau}{\partial \omega} > 0$ reflects task-specific differences in fragmentation costs attributable to these inherent differences. This attribute of the cost function is similar to Grossman and Rossi-Hansberg (2008). η_i captures producer i's information technology, while η_s reflects the state of technology in the sourcing location. I assume technology lowers fragmentation costs so that $\frac{\partial \tau}{\partial \eta} < 0$. ρ_j represents the extent to which production technology in industry j is amenable to electronic communication.

If electronic communication about the production process lowers fragmentation costs, then $\frac{\partial^2 \tau}{\partial \eta \partial \rho} < 0$. Intuitively, a firm's communication technology will have a bigger impact on costs when its production process can be codified in an electronic format. A sourcing location's technology may also have an impact on the effectiveness of firms' communication technology. If so, then we expect the cost-reducing effect of firm-level technology to increase in the sourcing location's technology ($\frac{\partial^2 \tau}{\partial \eta_i \partial \eta_s} < 0$ and $\frac{\partial^2 \tau}{\partial \rho_j \partial \eta_s} < 0$.)

Final good producers pay the task specific fragmentation costs in units of labor from sourcing location s. The per-unit cost to final good producer i for task k purchased

[18] In this setup, fragmentation lowers production costs only through a cheaper wage. In practice, MSPs also enjoy gains to specialization that provide an incentive for fragmentation even when wages are the same. The model can easily be extended to capture this by assuming MSPs require $\alpha < 1$ units of labor per task output.

from an MSP in location s is:

$$c_{kijs} = w_s \tau(\delta_{is}, \omega_k, \eta_i, \eta_s, \rho_j). \tag{2.4}$$

Fragmenting only maximizes variable profits if it results in lower costs of task production. Without loss of generality, order tasks such that fragmentation costs are strictly increasing in the index k for a given location. A necessary, though not sufficient, condition for fragmentation is then

$$w_h > w_D \tau_D(0) \text{ or} \tag{2.5a}$$
$$w_h > w_O \tau_O(0), \tag{2.5b}$$

where D and O denote the lowest cost domestic and offshore locations respectively, and $\tau(0)$ denotes the fragmentation cost of task $k = 0$. Equation (2.5) simply states that the task with the lowest fragmentation cost must be cheaper to fragment, either domestically or offshore, than to produce in an integrated plant. Whenever equation (2.5a) holds, then for offshoring to be potentially viable, it must also be the case that

$$\frac{w_O}{w_D} < \frac{\tau_D(0)}{\tau_O(0)}. \tag{2.6}$$

In this case, the decision to offshore is independent of the home wage and depends only on the relative costs and benefits of sourcing from the firm's lowest cost domestic location relative to its lowest cost foreign location.

Equations (2.5) and (2.6) highlight the role of relative wages and costs in determining whether fragmentation and offshoring take place. If the wage differential is not sufficiently high relative to fragmentation costs, then producers will not fragment and non-participation arises without any role for fixed costs and productivity.[19]

When a producer only sources from one location s, then its optimal share of fragmented production, $\bar{k}_s$, is implicitly defined by

$$w_h = w_s \tau_s(\bar{k}_s), \text{ where } s \in \{D, O\}. \tag{2.7}$$

[19]The other potential corner solution is $w_h > w_s \tau_s(1), \text{ where } s \in \{D, O\}$. In this case, producers fully fragment. Since the focus of this paper is on U.S. manufactures that still perform some fraction of their physical transformation activities, I assume w_h is sufficiently low relative to costs so that full fragmentation does not occur.

The cost of the composite input M, for producer i sourcing from s is then:

$$C_{is} = (1 - \bar{k}_s)w_h + w_s \int_0^{\bar{k}_s} \tau_{is}(k)\, dk, \text{ where } s \in \{D, O\}. \tag{2.8}$$

Figure 1a illustrates the case where offshoring maximizes variable profits by minimizing the cost of producing M.

This new cost for the composite input M results in the following profits for producer i:

$$\pi_{is} = \frac{(1-\sigma)A}{\sigma^{(1-\varepsilon)}} \left(\frac{\varphi}{C_{is}} \right)^{(\varepsilon-1)} - f_s, \text{ where } s \in \{D, O\}. \tag{2.9}$$

2.3 Role of productivity, local wages, technology, and distance

Producers choose the fragmentation strategy that maximizes profits. Of the firms for which fragmentation maximizes variable profits, fragmentation is only optimal for those with productivity above an endogenous threshold. Figure 1b illustrates the thresholds φ_D, above which domestic fragmentation is profitable, and $\varphi_O > \varphi_D$, above which offshoring is optimal. The option to fragment domestically increases φ_O, which translates into a higher opportunity cost of offshoring. The option to fragment domestically therefore increases the likelihood that an individual firm will fragment, but decreases the likelihood that it will do so offshore.

The relative slopes of the profit functions illustrated in Figure 1b depend upon the cost of the composite input M. Given the assumptions on the cost function $\tau()$, this implies that the relative profitability of fragmentation is i) increasing in a plant's local labor cost ($\frac{\partial \pi_I / \partial w_h}{\partial \pi_s / \partial w_h} > 1$); ii) decreasing in distance to potential suppliers ($\frac{\partial \pi_s}{\partial \delta} < 0$, while $\frac{\partial \pi_I}{\partial \delta} = 0$); and iii) increasing in firm communication technology ($\frac{\partial \pi_s}{\partial \eta} > 0$, while $\frac{\partial \pi_I}{\partial \eta} = 0$). In addition, the profitability of domestic versus foreign sourcing will depend on how each these factors affects the relative costs of each alternative. Formal derivations of the productivity thresholds and comparative statics on wages, technology, and distance are in the mathematical appendix.

A main contribution of the model is to provide a framework in which to consider how the impact of communication technology should vary across industries when it is used to lower coordination costs. In particular, we expect a larger impact on the

profitability of fragmentation when communication technology is used by firms in industries in which the production process is easier to codify in an electronic format. The model also provides an intuitive framework in which to interpret how suppliers' technology may interact with fragmenting firms' technology.

3 Empirical analysis of domestic and foreign fragmentation

The model predicts that a manufacturer's relative profits from each sourcing strategy depend upon its potential labor cost savings and fragmentation costs. Since I do not observe the potential costs for each sourcing alternative, I rely on the assumption that a plant chooses the option that maximizes its profits. I estimate the probability that a plant will fragment production, and if so, whether it does so domestically or offshore. I first estimate this probability via a nested logit model (NLM). This specification maps most directly from the theory by jointly estimating the probability of all three sourcing alternatives. Because a NLM relies on stringent assumptions about the error structure, I also estimate the probabilities of fragmentation and offshoring using a linear probability model (LPM). Both estimation techniques provide strong evidence of the same qualitative story.

The NLM allows for correlation of the errors across different sourcing alternatives. I estimate the NLM by grouping domestic and offshore fragmentation in the same nest to account for unobserved characteristics that make a plant's production process easier to fragment. This specification does not imply a sequential structure to a plant's sourcing decision, but instead relaxes the independence of irrelevant alternatives (IIA) assumption present in a multinomial logit estimation.

Figure 2 illustrates the nesting structure. Following the discrete choice literature, I assume that a plant's latent profit function is a linear function of plant and fragmentation location characteristics, and furthermore additively separable into a component that varies only with the decision to fragment and a component that reflects the profits derived from fragmenting domestically or offshore. Under these assumptions, the latent profit function for plant i can be written as

$$U_{ifs} = z'_{if}\alpha + w'_{ifs}\beta + \varepsilon_{ifs}, \tag{3.1}$$

where f denotes the level one decision to fragment or not and s denotes the plant's

level two fragmentation choice within each nest.[20] If the distribution of ε_{ifs} is given by a multivariate extreme value with parameter λ, then the probability that plant i chooses sourcing alternative s in nest f is given by:

$$\begin{aligned} P_{ifs} &= P_{if} \times P_{is|f} \\ &= \frac{e^{z'_{if}\alpha+\lambda_f I_{if}}}{\sum_{l=1}^{2} e^{z'_{if}\alpha+\lambda_l I_{il}}} \times \frac{e^{w'_{ifs}\beta/\lambda_f}}{\sum_{m\in f} e^{w'_{ims}\beta/\lambda_f}} \end{aligned}$$

where $I_{if} \equiv ln\left(\sum_{m\in f} e^{w'_{ifs}\beta/\lambda_f}\right)$ is the inclusive value for nest f.

Since the majority of variables available for the analysis are case-specific, for estimation purposes, it is useful to re-write equation 3.1, distinguishing between the case and alternative-specific variables.

$$U_{ifs} = z'_i\alpha_f + \beta_{dist} dist_{ifs} + x'_i\beta_{fs} + \varepsilon_{ifs}. \tag{3.2}$$

From the model, z includes a plant's local labor costs, $dist_{ifs}$ is a measure of distance between a plant and each fragmentation alternative, and x is a vector that includes plant-level productivity and communication technology.

3.1 Data on fragmentation costs and benefits

In the model, labor cost savings are a function of a plant's local wages relative to labor costs in each sourcing location. The fragmentation data do not provide information on plants' sourcing locations, so I capture the potential for labor cost savings using plants' local labor costs. I measure these costs with state-industry level wages constructed from six-digit production worker occupations from the Bureau of Labor Statistics' Occupation Employment Statistics. The mix of occupations only varies at the industry level so that variation in wages across states is solely attributable to differences in states' wages for detailed occupations. This wage measure ensures that variation in worker composition at the plant and state level does not drive the results.[21] Because wages are specific to a plant's location, rather than alternative-

[20] The no fragmentation nest is degenerate since plants that do not fragment production have no additional decision to make. To achieve identification in the estimation process, I therefore set the dissimilarity parameter to one for this nest.

[21] Skill differences across occupations are likely a much bigger problem than differences within occupations. For example, Handwerker and Spletzer (2010) find that the majority of wage differences within plants are driven by differences across occupations. Unfortunately, the occupation-level data are only available at the state level.

specific, the NLM requires that I specify the level in which to include it. In the model, a firm's decision to offshore depends on $w_D/w_O > \tau_O/\tau_D$, and is thus independent of w_h. The local wage should therefore only affect the level 1 decision to fragment or not. I also include four-digit NAICS fixed effects in level 1 to control for all possible industry-level characteristics that could affect a plant's fragmentation decision.

I construct an alternative-specific measure of the log of distance to potential suppliers. Distance to domestic suppliers is the distance between a plant and the closest manufacturing service provider (MSP). I identify MSPs as all manufacturing establishments that specify their primary activity as "Providing contract manufacturing services to others."[22] Fragmenting plants are an average of 1.6 miles away from an MSP compared to 2.2 miles for plants with no CMS purchases. I measure distance to a potential foreign sourcing location as the distance to the closest deep water port or border crossing with Canada or Mexico. Plants that offshore are an average of 50 miles closer to a foreign entry point than domestic sourcers. For no fragmentation, I set the distance equal to a small fraction, rather than zero, so that the log of distance is defined for each sourcing alternative.[23]

Plant communication technology is captured by an indicator equal to one if the plant used electronic networks to control or coordinate its shipments in 2007. 36 percent of non-fragmenters, 51 percent of domestic fragmenters, and 61 percent of offshorers used electronic networks for this purpose in 2007. Although this measure directly relates to shipments rather than input purchases, it is a useful proxy for whether a plant has integrated the ability to communicate electronically with its physical production process. Data from the 1999 Annual Survey of Manufactures (ASM) Computer Survey Network Use Supplement (CNUS) show that just over half of the manufacturing plants that used networks to coordinate shipments in 1999 also used networks to make input purchases (DOC, 2001). The CNUS data also show that 32 percent of plants that sold goods over networks also used networks to provide information about their design specifications to external suppliers, compared to only 16 percent of plants that did not sell goods over networks.[24] I include this case-specific variable in level 2 to allow

[22]The 2007 CM asked all plants in the CMS sample to identify their primary activity from four choices. "Providing contract manufacturing services to others" is one of the four options. The results are robust to using a weighted distance of a plant's input suppliers, using the 2002 BEA I-O tables to identify inputs and their respective weights in production. Additional information for both measures is in the data appendix.

[23]This assumption is not necessary in the linear probability model, since I cannot employ an alternative-specific measure in that framework and instead estimate a separate coefficient for each of the distance measures.

[24]Plants that do not use networks to control or coordinate shipments may still use the internet. For example, DOC (2001) finds that approximately 87 percent of manufacturing plants in the 1999 ASM sample used an electronic network at their plant. In contrast, only 31 percent and 33 percent

for a differential effect of technology on the conditional probability of offshoring. I also control for potential industry differences in the number of fragmentable tasks using the share of differentiated inputs in a plant's industry from Nunn (2007).[25]

An obvious problem with estimating equation (3.2) is that plant communication technology may be endogenous to its sourcing strategy. If a plant begins using electronic networks because it is fragmenting production, then an unobserved positive shock to fragmentation will bias the estimated coefficient on technology. To address this issue, I focus on a specific mechanism through which technology lowers fragmentation costs. In the model, technology lowers fragmentation costs by making it easier to communicate design criteria and production specifications across locations. In practice, a plant's use of electronic networks for these purposes depends upon its ability to codify the design and production requirements in an electronic format. If this ability varies across industries, electronic networks will lower fragmentation costs more in those industries in which the production process is amenable to electronic codification ($\frac{\partial^2 \tau}{\partial \eta \partial \rho} < 0$). Identifying a differential impact of electronic networks that varies systematically with industry-level codifiability provides evidence that electronic communication lowers fragmentation costs through this specific channel.[26]

I measure variation in industry electronic codifiability as the share of plants in a four-digit NAICS industry that used Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) software in 1999, calculated from the CNUS data. CAD/CAM software is the predominant tool used to translate designs and specifications into an electronic format. There is substantial variation in CAD/CAM use across industries. In "Other apparel" manufacturing only four percent of plants use CAD/CAM, while almost all plants use CAD/CAM in "semi-conductor machinery manufacturing." This variation is driven by the complexity and extent to which the physical transformation process can be codified electronically, and is largely exogenous at the plant level. Conversations with suppliers at CMS trade shows indicate that the ability to communicate electronically has lowered fragmentation costs more for fragmentation in CAD/CAM intensive industries relative to low CAD/CAM industries. Consistent with the premise that CAD/CAM facilitates communication about design specifications, the correlation coefficient between CAD/CAM intensity and the frac-

of the ASM plants accepted or placed orders online respectively.

[25] I use the liberal definition of the fraction of inputs not sold on an exchange and not reference priced.

[26] This approach is similar to the identification strategy in Rajan and Zingales (1998) who assess the effect of financial development on country growth by estimating the differential impact financial development has on growth in industries that are dependent on external financing. Those authors note "One way to make progress on causality is to focus on the details of theoretical mechanisms through which financial development affects economic growth, and document their working" (p.560).

tion of plants in an industry that use electronic networks to share product designs with suppliers is 0.5, with a p-value of 0.0.[27]

3.2 Results from the nested logit model

Table 4 presents the coefficients from estimating equation 3.2. Column 1 presents the baseline results. Since the wage measure only varies by state, the standard errors are clustered at this level.[28] All coefficients have the expected sign and are statistically significant. The negative coefficient for distance, the only alternative-specific regressor, implies that increasing the distance between a plant and a potential sourcing location decreases the probability that the plant will source from that location. The remaining variables are all case-specific, so their coefficients are interpreted relative to the omitted category, as in a multinomial logit model. The level 1 panel shows that plants in high wage states are more likely to fragment production. The level 2 panel estimates suggest that more productive plants and those using electronic networks are more likely to fragment domestically or offshore than to produce in-house. Column 2 presents results from adding the interaction between plant technology and industry CAD/CAM intensity.

The estimated wage coefficient will be biased if state-level wages are correlated with differences in worker skill or local demand and these factors affect fragmentation. Another potential issue is that a high concentration of manufacturing employment in a state may lead to both more fragmentation and to higher wages. Columns 3-5 show that the estimated coefficients and their statistical significance are all robust to controlling for i) state-level skill differences, measured using the American Community Survey data as the share of workers with a college degree; ii) variation in local demand, measured as personal income in a plant's local market from the Bureau of Economic Analysis; and iii) manufacturing concentration, measured as the share of manufacturing employment to population in a state.[29] Column 6 addresses the possibility that location may be endogenous to a plant's fragmentation strategy, which could bias estimates on the effect of distance, by using the subset of plants that have existed in the same physical location for at least ten years. Once again, the coefficients and their significance are largely unchanged. The dissimilarity parameter

[27] Additional details on the CAD/CAM measure are in the data appendix.

[28] Clustering by industry does not change the statistical significance of the coefficients.

[29] I have also used measures of MSP concentration near the plant. While this does affect the estimate on the distance to the closest MSP, the wage coefficient is robust. These results are available upon request.

for the fragmentation nest, λ_F, is about 0.2 in all specifications. This estimate is statistically different from one, confirming that IIA is violated within the fragmentation nest and that a NLM is therefore appropriate.

3.3 Economic interpretation of the estimates

As is well known, the coefficients from a non-linear model must be transformed in order to assess their marginal effects. Table 5 provides the average marginal effects (AMEs) and their standard deviations for the main variables of interest. I do not calculate a statistical significance for each effect, and instead rely on the standard errors reported in Table 4 which Greene (2010) argues are both a sufficient and more informative metric. The top panel of Table 5 shows that doubling the distance to potential domestic suppliers or a foreign entry points decreases the likelihood of fragmentation by about one percentage point. Estimates for the conditional probability of offshoring, reported in the last column, indicate that doubling the distance to potential domestic suppliers increases the likelihood of offshoring by 2.3 points. As expected, the probability of fragmentation is increasing in plants' local wage. The AME for wages implies that a plant in Washington state is almost four percentage points more likely to fragment than an otherwise identical plant in Mississippi, where average production worker wages are almost 30 percent lower. Since this variable is estimated in level 1, it is constrained to have no impact on the conditional probability of offshoring. Consistent with the fixed cost assumptions in the model, both the probability of fragmentation and the conditional probability of offshoring are increasing in plant productivity.

Ai and Norton (2003) show that the standard transformation for marginal effects does not provide the correct interpretation for an interaction term in a non-linear model. In more recent work, Greene (2010) argues that figures provide the most informative economic interpretation of interaction terms in these models. Following this approach, Figure 3 illustrates the AMEs of electronic communication by industry CAD/CAM intensity for the probability of each outcome. Panel (a) shows that plants using electronic networks are more likely to fragment, and that this is particularly true for plants in CAD-intensive industries. Panel (b) depicts a positive relationship between electronic network use and the probability of domestic fragmentation that is almost twice the size for plants in the most CAD-intensive industries relative to those in the least CAD-intensive industries.

Panels (a) and (b) in Figure 3 provide evidence that electronic communication facil-

itates fragmentation by lowering the costs of communicating design and production specifications across locations. In contrast, panel (c) suggests that this mechanism is NOT present for fragmentation offshore. While use of electronic networks does have a positive relationship with the probability most plants will offshore, its impact is decreasing in CAD intensity and negative for plants in the most CAD-intensive industries. Panel (d) highlights the differential impact across industries showing that the conditional probability of offshoring is strongly decreasing in CAD intensity.

3.4 Estimation using a linear probability model

The NLM relies on stringent assumptions about the error structure and proportionate impact of variables within and across nests. To ensure that the results are not an artifact of these assumptions, I estimate linear probability models (LPMs) for the probability of fragmentation, and for the probability of offshoring conditional on fragmentation. Although the LPM does not allow for joint estimation of all three options, it provides a useful complement to the NLM estimates by relaxing the strong assumptions on the errors, allowing for detailed six-digit NAICS fixed effects, and estimating separate coefficients for both levels that are easily interpretable. I estimate:

$$Pr(y_{i,j,h} = 1|X_{i,j,h}) = \beta_J + \beta_T Tech_i + \beta_w wage_h + \sum \beta_D Dist_i + \sum \beta_P Prod_i, \quad (3.3)$$

where $y_{i,j,h}$ equals one if plant i in industry j and home state h purchases CMS.

The technology and wage variables are the same as those used in the NLM. β_J are six-digit NAICS fixed effects. Since the OLS framework allows for more detailed industry controls, I now use a six-digit NAICS measure of industry CAD/CAM intensity. OLS estimation does not allow for alternative-specific regressors, so I include the domestic and offshore measures separately. This allows me to create indicators for distance categories, which is desirable given the non-linear impact of distance on domestic shipments documented in Hillberry and Hummels (2008). I classify plants into productivity terciles to follow more closely the model's prediction that producers must exceed a productivity threshold for fragmentation to maximize profits.[30] I also estimate a variant of equation 3.3 for the subset of fragmenting plants, where $y_{i,j,h}$ equals one if plant i in industry j and home state h purchases CMS offshore.

As a further step towards addressing the potential for reverse causality between tech-

[30]The NLM included continuous rather than discrete measures to minimize convergence and estimation issues that arise with fixed effects in non-linear models.

nology and fragmentation, the LPM framework also allows for an instrumental variable (IV) estimation. An IV is also useful here since the electronic network indicator is a proxy variable that is therefore likely to have some measurement error. I implement the IV using each plant's 2002 use of electronic networks and productivity as instruments. The share of plant using networks to control or coordinate shipments in 2002 is roughly half the 2007 share for each CMS category. For the lagged instrument to be valid, plants' use of networks in 2002 must be determined by factors other than their decision to fragment production in 2007. For example, the instrument can identify a causal relationship if plants that used networks in 2002 to facilitate sales decide to fragment in 2007 because their existing communication technology made fragmentation relatively more profitable.

There are three reasons to believe that factors other than fragmentation play a role in plants' use of networks to control shipments: 1) almost half of the plants that used networks to control shipments in 1999 did not use networks to purchase inputs (DOC, 2001); 2) McElheran (2010) finds that, although E-buying and E-selling share the same technology platforms, E-selling generally entails more complex organizational changes; and 3) in 2007, one third of plants that did not fragment production used electronic networks. The instrument's power to identify a causal relationship also depends upon the existence of plants that used networks in 2002 but did not fragment production. Although the 2002 CM did not ask the same 2007 CMS purchase question, it did ask whether a plant purchased CMS from another firm. This question allows me to identify, for all single-unit firms, whether or not the firm purchased CMS in 2002.[31] I therefore check my results using the subset of single unit firms that DID NOT fragment in 2002 to ensure that plants' 2002 technology status is not a function of their current fragmentation choice.

3.4.1 Results from the linear probability model

Table 6 presents the LPM results. The left panel provides estimates for the probability of fragmentation and the right panel corresponds to the probability of offshoring, conditional on fragmentation. The OLS estimates indicate that plants using electronic networks are ten percentage points more likely to fragment production, and conditional on fragmentation, two points more likely to offshore. The probabilities

[31] A single unit firm that fragments production must do so by contracting with another firm. A plant that belongs to a multi-unit firm could purchase CMS from another plant in the firm. This analysis also shows that the results are not driven by differences between intra-firm and outsourced fragmentation.

of fragmentation and offshoring are both increasing in plant productivity, with a stronger impact for plants in the top productivity tercile. These estimates are all reassuringly similar to the average marginal effects from the NLM.

OLS estimates for the effect of local wages on the probability of fragmentation also closely match the nested logit average marginal effects. A 30 percent increase in the wage is associated with a five point increase in the probability of fragmentation. In contrast to the NLM, the LPM estimates imply that, conditional on fragmentation, plants in low wage states are more likely to offshore. By including plants' local wage in level 1, the NLM restricts the conditional probability of the wage to be zero. While this is consistent with the model's predictions, the OLS results suggest that it is incorrect. A simple extension of the model can rationalize this finding. Suppose there is one high wage domestic state, one low wage state and one offshore location, where $w_O < w_L < w_H$. For simplicity, assume that $\tau_O(.) = \tau_L(.) = \tau_H(.)$ so that $w_O\tau_O(.) < w_L\tau_L(.) < w_H\tau_H(.)$. With this set-up, offshoring maximizes variable profits for all plants, but only those producers with productivity above $\tilde{\varphi}_O$ will source offshore. While plants in the high wage state with $\tilde{\varphi}_D < \varphi < \tilde{\varphi}_O$ will fragment domestically, no producers in the low wage state will do so. Intuitively, firms in the lowest wage states must source offshore to access cheaper labor.

The OLS estimates also confirm an important role for distance in sourcing strategies. Relative to plants with a domestic supplier within a five mile radius, plants 5-20 miles away from potential domestic suppliers are 2.2 percentage points less likely to fragment, and those over 20 miles away are 2.9 points less likely to fragment. Plants that are over 200 miles away from a deep water port are no more likely to fragment, but conditional on fragmentation, are 1.6 points more likely to offshore. The estimates also show that plants over 50 miles from a border crossing with Mexico are almost two points less likely to fragment and over ten points less likely to offshore, conditional on fragmentation.

The IV estimates depict an even stronger relationship between plant communication technology and fragmentation, suggesting that they help mitigate measurement error from using a proxy variable for communication technology. The "Adopters" column in each panel shows that the IV estimates are generally robust when the sample is limited to plants that did NOT fragment production in 2002. First stage regressions are presented in Table 8. The lagged instrument is always significant in its own first stage regression, with the expected positive coefficient. In addition, the F-statistics for the null hypothesis that the instruments are jointly equal to zero are well above the threshold of 10 proposed in Stock, Wright and Yogo (2002).

IV estimation on the subset of "adopters" ensures that plants' use of networks in 2002 is not a function of their current fragmentation status, but it cannot address the potential that a plant used networks in 2002 because it planned to fragment in the future. The last column in each panel in Table 6 therefore presents estimates in which plant use of networks is interacted with industry CAD/CAM intensity. The estimated coefficients are similar to the NLM marginal effects. Figure 4a plots the full effect of electronic networks on the probability of fragmentation evaluated at different levels of CAD intensity. Relative to plants that do not use electronic networks, plants that use networks in "other apparel" manufacturing (CAD intensity of 4 percent) are only 4.5 percentage points more likely to fragment, while plants using networks in the "semiconductor machinery" manufacturing (CAD intensity of 98 percent) are over 30 points more likely to fragment than plants not using networks. Estimates for the conditional probability of offshoring also match the NLM results. Figure 4b illustrates that fragmenting plants using networks in the most CAD-intensive industries are less likely source offshore. The data appendix shows that the LPM results are also robust to i) state-level skill differences; ii) variation in local demand; iii) manufacturing concentration; and iv) using the subset of plants that have existed in the same physical location for at least ten years.

Results from both the NLM and LPM estimates support the premise that communication technology lowers fragmentation costs by facilitating communication about design and production specifications. This mechanism is not evident, however, in plants' decisions to locate fragmented production offshore. Instead, both the NLM and the LPM estimates show that plants using networks in CAD/CAM intensive industries are less likely to offshore. A possible explanation for this result is that the state of technology in offshoring locations is below the U.S. level. Receiving CAD/CAM files and using them correctly requires sophisticated equipment and workers. As a result, CAD/CAM's potential to reduce coordination and communication costs across locations when used in conjunction with electronic networks cannot be fully realized if a given location does not have the ability to use it. In the next section, I exploit the firm-country import data to assess this hypothesis.

3.5 Location of offshore production

The linked import data provide information on firms' decisions about where to locate their offshored production. I aggregate the import data to the firm-country level and construct an indicator equal to one if a firm sources from a given country. The dataset therefore includes an observation for every potential firm-country import combination.

I regress the indicator on firm characteristics (X_F), country characteristics (X_C), and firm-country interactions ($X_F \times X_C$):

$$Pr(y_{i,c} = 1|X_{i,c}) = \theta + \beta_F X_F + \beta_S X_C + \beta_{FC}(X_F \times X_C), \tag{3.4}$$

where $y_{i,c} = 1$ if firm i imports from country c. The firm variables, X_F, include the log of value added productivity, electronic network use, the interaction between electronic networks and industry CAD intensity, and the industry share of differentiated inputs.[32] The country variables, X_F, include relative wages, human capital, and a country technology measure. I interact the firm-level technology variables with country technology to assess whether the impact of firm technology on the probability of sourcing from a given location depends upon that country's level of technology. The interaction terms, $X_F \times X_C$, also include the minimum distance between each firm and country.

Country technology is measured as the number of secure internet servers in 2007 in each country from the World Bank World Development Indicators. This variable is similar to one used by Freund and Weinhold (2002), who find an important role for a country's internet penetration in explaining variation in countries' growth of U.S. services trade. The number of internet servers in a country represents a measure of countries' communication technology infrastructure. In this dimension, an assessment of its importance in trade complements Limão and Venables (1999), who show that transportation infrastructure is a significant determinant of trade flows. I calculate the minimum distance between each firm and each country using latitudes and longitudes from the CEPII.[33] The relative foreign wage is constructed from country-industry-occupation wages from the International Labor Organization.[34] Because the model assumes homogeneous labor, I control for country human capital with an updated version of the Hall and Jones (1999) measure.[35] Multi-unit firms may span multiple industries, so I control for industry using the firm's share of employment in each four-digit NAICS code.

Since the objective of this section is to assess whether the effectiveness of firm-level technology depends upon the level of technology in a given country, I use country

[32]Since the focus of this analysis is on the interaction terms, I use a continuous productivity measure to facilitate the disclosure analysis.

[33]The data are available here: `www.cepii.fr/anglaisgraph/bdd/distances.htm`. I use latitude and longitude for the largest city in each country. Firm latitude and longitude is based on the closest manufacturing plant in the firm.

[34]See Oostendorp (2005) for details about the data. The wage construction methodology is described in the data appendix.

[35]I use the education data from Barro and Lee (2000) to construct a measure of human capital.

or firm fixed effects and focus on the interaction terms. This approach controls for all possible omitted country or firm characteristics and is therefore less susceptible to reverse causality and spurious correlation problems. I estimate equation (3.4) via OLS.[36]

Table 7 presents results from estimating equation (3.4) on the subset of firms that primarily purchase CMS from offshore and domestic locations respectively. Columns 1 and 2 report estimates using country fixed effects. This specification allows for a quantification of the role of firm technology in the probability of sourcing from a given country, while controlling for all possible omitted factors that vary at the country level. The interactions between firm and country technology are positive and significant in both samples. Figure 5a plots the full effect of electronic network use for offshoring firms evaluated at different levels of country technology. While using electronic networks has a zero or negative effect on the probability of sourcing from countries with a small number of servers, such as Bangladesh, their use increases the probability of sourcing from a high technology country like Japan by over two percentage points.

Figure 5b depicts a similar relationship between industry CAD intensity and country technology. To illustrate magnitudes, increasing CAD intensity from its level in the "other apparel" industry to its level in the "semi-conductor machinery" industry is associated with no increase in the probability of sourcing from a country with Bangladesh's technology, but a 6.6 percentage point increase in the probability of sourcing from a country with Japan's level of technology. Columns 3 and 4 in Table 7 report estimates of equation (3.4) with firm fixed effects. The firm-country interactions are positive and statistically significant in this specification as well. The probability a firm will source from a given country is increasing in the number of servers in that country, but the effect of servers is larger for firms that use electronic networks and is increasing in industry CAD intensity. To illustrate the implied magnitudes of the coefficients, an increase in country technology equivalent to increasing Bangladesh's number of servers to the level in Japan is associated with a 14 percentage point increase in the probability a firm in the "other apparel" industry sources from that country, and a 25 percentage point increase for a firm in the "semi-conductor machinery" industry.

[36]OLS is preferable in this context due to the inclusion of firm and country fixed effects. As discussed in the NLM section, the marginal effects for interaction terms vary by observation making them difficult to summarize. Finally, both probit and logit models suffer from perfect separation that occurs when a firm or country dummy predicts an outcome perfectly. There is no appealing solution for the separation problem in the analysis here with fixed effects. See Zorn (2005) for a discussion of this issue.

Although the specifications with country fixed effects control for all other factors that vary at the country level, the interaction coefficients rely on a measure of country technology for identification. To the extent that country technology is correlated with other factors, these results cannot definitively pin down the specific mechanism through which the impact of firm technology depends upon country characteristics. The results therefore provide strong evidence that firm technology depends upon supplying countries' characteristics, but do not rule out the potential that factors correlated with country technology (such as human capital or intellectual property rights protection) may be equally or more important than technology itself.

The results presented in Table 7 are also consistent with an important role for distance. The estimates with country fixed effects, presented in columns 1 and 2, exploit differences in firms' distance to a given country that arise from variation in firms' geographic distribution across the U.S. They suggest that doubling the firm-specific distance to a country is associated with 2.6 (1.9) percentage point decrease in the probability of sourcing from that country for offshoring (domestic) fragmenters. The estimates with firm fixed effects exploit the variation in distance to different countries for a given firm. Although the coefficients have the expected negative sign, they are not statistically significant.

4 Assessing the relative importance of technology, distance and wages

Before concluding, it is useful to assess the relative importance of wages, technology, and distance in firms' fragmentation and offshoring decisions. To perform the analysis, I calculate the share of the explained variation attributable to each of these key determinants. For example, to assess the importance of wage variation in a given estimation, I first calculate $\hat{y}$, the predicted value of the dependent variable for each observation, using actual values of independent variables. Next, I re-calculate predicted values using the actual wage values for each observation, but holding all other independent variables at their sample means. Letting $\hat{y}_{iw}$ denote this predicted value for observation i, the share of the explained variation accounted for by the actual wage variation in the data is then the sample analogue of $VAR(\hat{y}_{iw})/VAR(\hat{y}_i)$. When multiple variables capture the effect of distance or technology, I calculate the predicted values with all distance or technology variables evaluated at actual values and all other independent variables held at their means. Since these calculations do

not incorporate covariances between independent variables, they do not represent a perfect decomposition and the fractions of the variation explained by different variables will not necessarily add to one. However, the analysis provides an informative quantification of the relative importance of the key explanatory variables.

Table 9 presents results of the simplified variance decomposition for the main results from the LPM estimates. The left panel corresponds to the IV estimates for the probability a plant will fragment production, and the right panel corresponds to the probability it will offshore, conditional on fragmentation. The first row reports results for the baseline specification and shows that wage variation accounts for one and two percent of the explained variation in fragmentation and offshoring decisions respectively. Technology, measured by plant use of electronic networks, accounts for 28 percent of the explained variation in plants' fragmentation decision and 24 percent of the explained variation in offshoring. Distance explains only one percent of the variation in the fragmentation decision, and two percent of the explained variation of offshoring. The second row shows that when the effect of electronic networks is allowed to vary by CAD intensity, technology accounts for 36 and 47 percent of the explained variation in fragmentation and offshoring respectively. Variation in communication technology accounts for much larger shares of the explained variation in plants' decisions to break up their production process and source from overseas, relative to distance and labor cost savings combined.

5 Conclusion

This paper documents the importance of domestic fragmentation of customized inputs in firms' global sourcing strategies. To my knowledge, it is the first analysis that covers firms' domestic and foreign sourcing decisions, and shows that domestic fragmentation is far more prevalent than offshoring, both within and across manufacturing industries. In addition, estimates from nested logit and linear probability models imply that, while electronic communication facilitates domestic fragmentation, it does not increase the likelihood of offshoring for all firms. Firms that have integrated electronic communication with their production process are less likely to offshore if they belong to an IT-intensive industry. These findings support the premise that a firm's communication technology lowers fragmentation costs by facilitating coordination across locations, but suggest that its effectiveness depends on suppliers' technological capabilities.

Firm-country location selection regressions support the hypothesis that the effect of firm technology on reducing offshoring costs is a function of its suppliers' technology. Firms using electronic networks, and those in IT-intensive industries, are more likely to source from high-technology countries. These results stand in stark contrast to many of the current assumptions about the role of technology in offshoring, and provide suggestive evidence that firms do not transfer their technological capabilities to low-cost sourcing locations. They also have different implications about how future technological developments affect global supply chains and production networks. Rather than benefiting low-wage countries, technological advances are more likely to result in U.S. firms sourcing from other high-tech countries (as in Grossman and Rossi-Hansberg, 2012), or to reshuffling their employment across U.S. states.

The empirical results show that labor cost differences and distance to potential suppliers are also important factors in a firm's extensive margin decisions about whether to fragment production, and if so, whether to offshore. However, a simple variance decomposition indicates that communication technology is more important than both of these factors combined. A fruitful avenue for future work may be to exploit the import data to assess whether this result holds on the intensive margin as well. The findings here also suggest that skill differences across industries and countries play a significant role in firms' sourcing decisions. Another direction for future work is to explore the effect of these differences, as well as the extent to which firms may transfer knowledge to the foreign locations from which they source.

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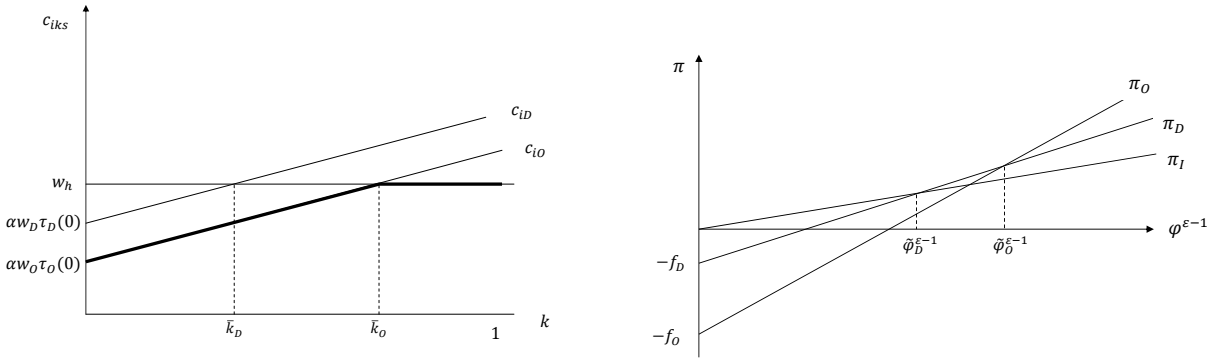
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Figure 1: Integrated production, domestic fragmentation or offshoring



(a) Variable costs per task

(b) Profits

Figure 2: Nested logit structure

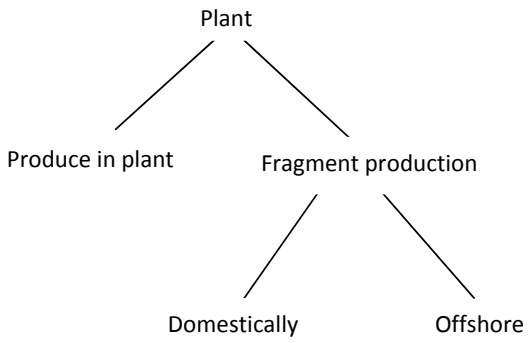
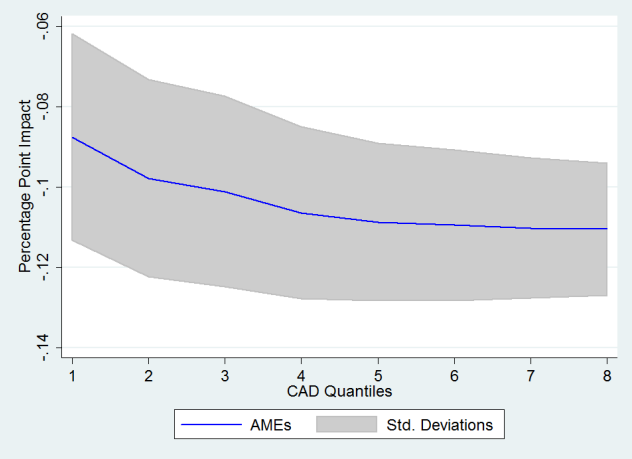
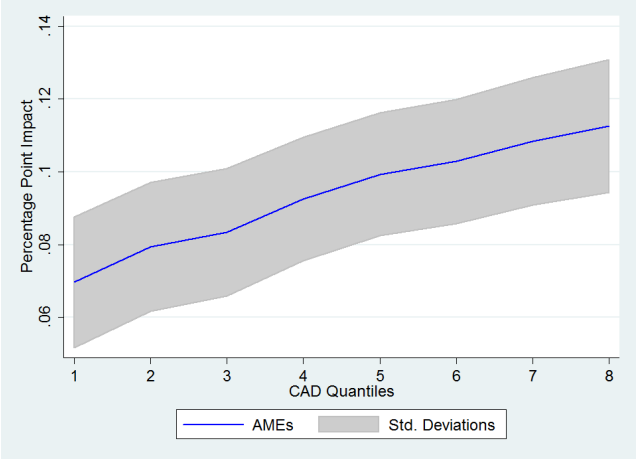


Figure 3: Average marginal effects of electronic networks, by industry CAD intensity

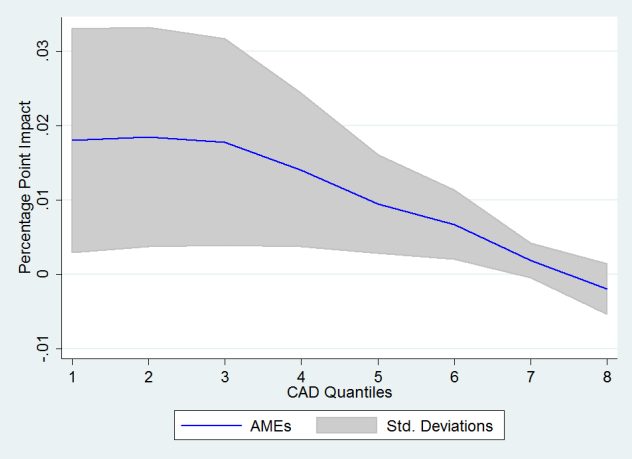
(a) No Fragmentation



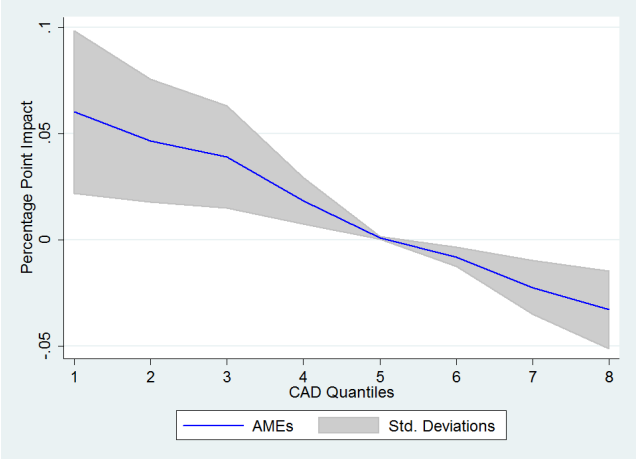
(b) Domestic Fragmentation



(c) Offshoring

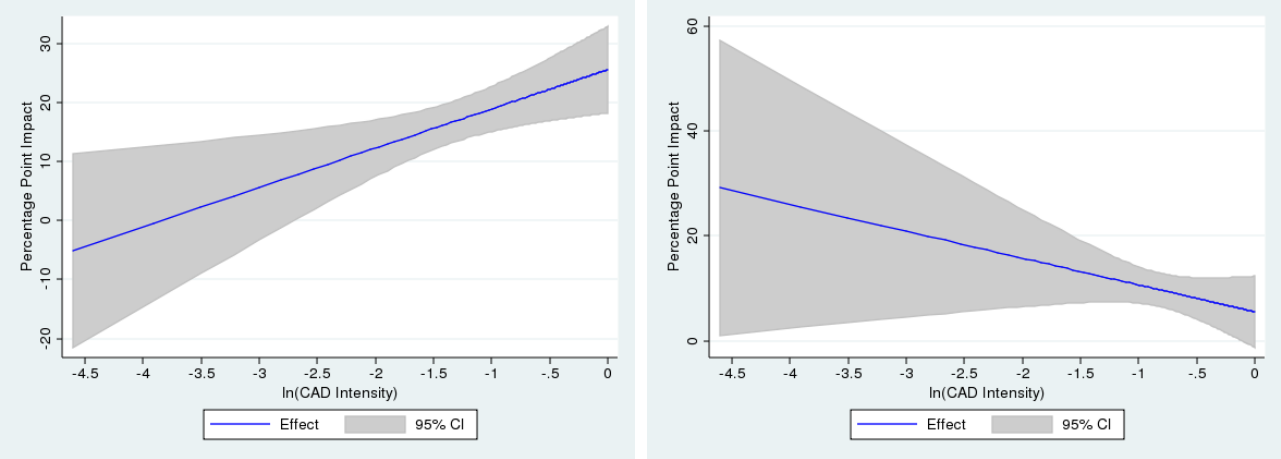


(d) Offshoring | Fragmentation



Notes: Average marginal effects of plant use of electronic networks evaluated at different levels of industry CAD/CAM intensity. Based on the estimates reported in column 2 of Table 4.

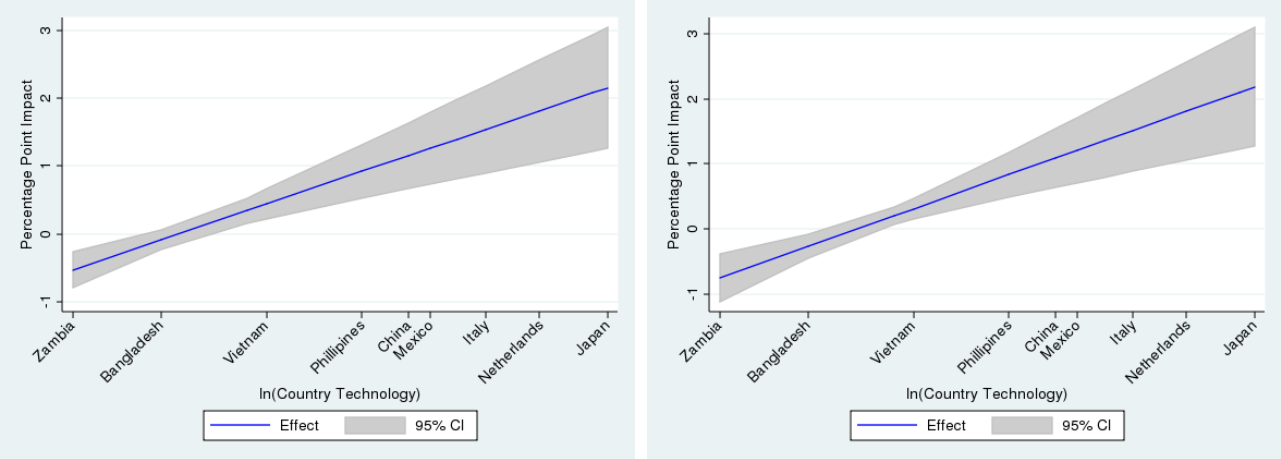
Figure 4: Differential impact of electronic networks, by industry CAD intensity



(a) Fragmentation (b) Offshoring | Fragmentation

Notes: Effect of plant use of electronic networks evaluated at different levels of industry CAD/CAM intensity. Based on the OLS estimates reported in column 4 of Table 6.

Figure 5: Differential impact of firm technology on probability of sourcing from country s, by country technology



(a) Electronic Network Use (b) Industry CAD Intensity

Notes: Effect of firm use of electronic networks and firm's average industry CAD/CAM intensity, evaluated at different levels of country technology. Based on the estimates with country fixed effects reported in column 1 of Table 7.

Table 1: Participation shares by contract manufacturing services purchase status

	Plant Shares			Firm Shares		
	Plants	Sales	Emp	Firms	Sales	Emp
No Purchases	0.71	0.57	0.61	0.69	0.31	0.42
Domestic Purchases	0.27	0.39	0.35	0.28	0.42	0.39
Offshore Purchases	0.02	0.04	0.04	0.02	0.03	0.03
Domestic & Offshore	na	na	na	0.01	0.24	0.16

Notes: Sales and employment shares weighted by the inverse probability of inclusion in the CMS sample.

Table 2: Plant means by contract manufacturing services purchase status

	Raw Means			Relative Ind. Means		
	Sales[a]	Emp	ln(VAP)	Sales	Emp	ln(VAP)
No Purchases	19,487	51.3	4.51	0.87	0.91	-0.03
Domestic Purchases	37,077	79.8	4.63	1.28	1.20	0.07
Offshore Purchases	51,457	137	4.74	2.17	1.69	0.20
All Plants	24,686	60.4	4.55	1.00	1.00	0.00

Notes: Relative ind. are means of plant values divided by NAICS 6 industry mean for sales and employment, and log differences for value-added labor productivity. All means weighted by the inverse probability of inclusion in the CMS sample. [a] Sales in $000s.

Table 3: Industry distribution of the share of establishments that purchase CMS

86 Manufacturing Industries (NAICS 4)

Domestic Purchases (% of estabs in industry)	Offshore Purchases (% of estabs in industry)				
	0%	0-5%	5-10%	10-20%	Total
5-10%	0	2	0	0	2
10-20%	1	22	2	0	25
20-35%	1	31	6	2	40
35-50%	0	13	4	1	18
50-60%	0	1	0	0	1
Total	2	69	12	3	86

Categories defined such that $LHS < \%estabs \leq RHS$

Table 4: Nested Logit Estimation Results

Dependent variable denotes plant i, in industry j and state h, fragmentation status, s

	Baseline	W/CAD	Demand	Skill	Manuf	10+ Yrs.
$ln(Distance_{is})$	-0.063***	-0.035***	-0.033***	-0.033***	-0.038***	-0.025***
	(0.006)	(0.008)	(0.007)	(0.008)	(0.008)	(0.009)
Level 1: Fragment						
$ln(wage_h)$	0.823***	1.182***	1.159***	1.088***	1.080***	1.227***
	(0.121)	(0.165)	(0.163)	(0.165)	(0.183)	(0.148)
$ln(BEA\ Income)$			0.008			
			(0.009)			
$ln(Share\ college_h)$				0.126		
				(0.089)		
$ln(Share\ manuf_h)$					0.078**	
					(0.037)	
NAICS 4 controls	yes	yes	yes	yes	yes	yes
Level 2: Domestic						
$Elec.\ networks_i$	0.524***	0.492***	0.493***	0.493***	0.489***	0.502***
	(0.014)	(0.028)	(0.028)	(0.027)	(0.028)	(0.033)
$\times ln(CAD_j)$		-0.029	-0.029	-0.029	-0.029	-0.011
		(0.029)	(0.028)	(0.028)	(0.029)	(0.030)
$ln(VAProd_i)$	0.211***	0.212***	0.211***	0.212***	0.212***	0.230***
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.016)
$ln(Diff\ inputs_j)$	0.266***	0.257***	0.257***	0.257***	0.256***	0.252***
	(0.045)	(0.046)	(0.046)	(0.046)	(0.046)	(0.052)
$ln(CAD_j)$		1.141***	1.232***	1.008***	0.854**	1.452***
		(0.290)	(0.291)	(0.300)	(0.352)	(0.293)
Level 2: Offshore						
$Elec.\ networks_i$	0.538***	0.431***	0.435***	0.435***	0.423***	0.458***
	(0.017)	(0.030)	(0.030)	(0.029)	(0.029)	(0.033)
$\times ln(CAD_j)$		-0.132***	-0.126***	-0.126***	-0.138***	-0.082**
		(0.036)	(0.036)	(0.035)	(0.036)	(0.036)
$ln(VAProd_i)$	0.247***	0.241***	0.239***	0.239***	0.243***	0.253***
	(0.015)	(0.016)	(0.016)	(0.016)	(0.016)	(0.019)
$ln(Diff\ inputs_j)$	0.580***	0.420***	0.410***	0.410***	0.431***	0.382***
	(0.072)	(0.072)	(0.069)	(0.070)	(0.073)	(0.077)
$ln(CAD_j)$		1.248***	1.331***	1.108***	0.968***	1.521***
		(0.274)	(0.276)	(0.284)	(0.338)	(0.275)
λ_F	0.182***	0.132***	0.124***	0.124***	0.142***	0.096***
	(0.021)	(0.031)	(0.029)	(0.030)	(0.030)	(0.034)
Log Likelihood	-67349.44	-67194.29	-67193.46	-67191.25	-67187.29	-48827.46
N	315,800	315,800	315,800	315,800	315,800	226,600

Notes: Demand controls for personal income in the plant's BEA Economic Area. Skill controls for the share of workers with a college degree. Manuf controls for the share of the population employed in manufacturing. 10+ years is the sub-sample of plants that have existed in the same physical location for at least 10 years. $ln(Diff\ inputs_j)$ is the share of differentiated inputs in an industry. Standard errors clustered by state. *, **, *** denote 10%, 5% and 1% significance respectively. N rounded for disclosure avoidance.

Table 5: Marginal Effects for Nested Logit Estimation

	ME on Probability of			
	None	Domestic	Offshore	Off | Frag
A. Baseline Specification				
$ln(Distance_{is})$				
Domestic	0.011	-0.017	0.007	0.023
	0.003	*0.008*	*0.006*	*0.016*
Offshore	0.010	-0.009	-0.002	-0.003
	0.003	*0.002*	*0.002*	*0.002*
$ln(wage_h)$	-0.156	0.143	0.013	
	0.044	*0.040*	*0.012*	
$Elec.\ networks_i$	-0.100	0.090	0.010	0.005
	0.028	*0.025*	*0.009*	*0.004*
$ln(VAProd_i)$	-0.041	0.033	0.008	0.013
	0.012	*0.010*	*0.007*	*0.009*
B. Baseline Specification, with CAD Interaction				
$ln(Distance_{is})$				
Domestic	0.006	-0.011	0.005	0.017
	0.002	*0.006*	*0.005*	*0.011*
Offshore	0.006	-0.005	-0.001	-0.002
	0.002	*0.001*	*0.001*	*0.001*
$ln(wage_h)$	-0.225	0.207	0.017	
	0.064	*0.056*	*0.015*	
$Elec.\ networks_i$	-0.092	0.095	-0.003	-0.029
	0.026	*0.028*	*0.004*	*0.019*
$ln(VAProd_i)$	-0.041	0.033	0.008	0.014
	0.012	*0.009*	*0.007*	*0.009*

Note: Average marginal effects (AMEs) for the Nested Logit specifications reported in Table 4. Standard deviations of AMEs reported in italics.

Table 6: Linear Probability Model Estimation Results

	Probability of Fragmenation				Probability of Off|Frag			
	OLS	IV Estimates			OLS	IV Estimates		
	Baseline	Baseline	Adopters	W/CAD	Baseline	Baseline	Adopters	W/CAD
$Elec.\ networks_i$	0.103***	0.183***	0.126***	0.256***	0.021***	0.100***	0.150***	0.055
	(0.004)	(0.019)	(0.034)	(0.037)	(0.003)	(0.016)	(0.038)	(0.035)
$\times ln(CAD_j)$				0.067***				-0.051
				(0.025)				(0.038)
$ln(wage_h)$	0.212***	0.170***	0.124***	0.162***	-0.116***	-0.125***	-0.177**	-0.122***
	(0.031)	(0.029)	(0.039)	(0.030)	(0.043)	(0.044)	(0.075)	(0.043)
$ln(VAProd_i)$ Q2	0.037***	0.117***	0.074*	0.121***	0.006	-0.001	-0.011	-0.003
	(0.005)	(0.023)	(0.039)	(0.023)	(0.004)	(0.018)	(0.049)	(0.019)
Q3	0.073***	0.155***	0.061***	0.158***	0.021**	0.038***	0.052*	0.036***
	(0.008)	(0.014)	(0.021)	(0.013)	(0.009)	(0.011)	(0.028)	(0.011)
MSP 5-20 miles away	-0.022***	-0.022***	-0.021***	-0.021***	0.000	0.001	-0.006	0.000
	(0.006)	(0.004)	(0.006)	(0.004)	(0.004)	(0.005)	(0.010)	(0.005)
20+ miles away	-0.029***	-0.028***	-0.020*	-0.028***	0.001	0.002	0.008	0.002
	(0.007)	(0.006)	(0.011)	(0.006)	(0.005)	(0.006)	(0.013)	(0.006)
Port is 51-200 miles	0.001	-0.006	0.010	-0.006	-0.007	-0.012	-0.012	-0.012
	(0.008)	(0.004)	(0.009)	(0.004)	(0.008)	(0.008)	(0.012)	(0.008)
200+ miles away	0.014	0.009*	0.017**	0.008	-0.016***	-0.023***	-0.024***	-0.024***
	(0.009)	(0.005)	(0.007)	(0.005)	(0.005)	(0.005)	(0.007)	(0.006)
50+ miles to Mexico	-0.019**	-0.034***	-0.097***	-0.033***	-0.123***	-0.094***	-0.091***	-0.095***
	(0.009)	(0.005)	(0.011)	(0.005)	(0.036)	(0.017)	(0.030)	(0.017)
50+ miles to Canada	-0.010	-0.006	-0.006	-0.006	-0.011*	-0.010	-0.009	-0.010
	(0.024)	(0.008)	(0.009)	(0.009)	(0.005)	(0.007)	(0.014)	(0.007)
NAICS 6 controls	yes	yes	yes	yes				
Adj. R-Squared	0.09	0.08	0.07	0.08	0.10	0.08	0.03	0.07
N	71,600	71,600	22,000	71,600	21,500	21,500	5,000	21,500

Notes: IV regressions instrument for electronic networks and productivity using lagged 2002 values. Standard errors clustered by state. *, **, *** denote 10%, 5% and 1% significance respectively. N rounded for disclosure avoidance.

Table 7: Offshore sourcing location selection

Dependent variable is 1 if firm f, in industry j, imports from country c

	Firms' CMS purchases are primarily:			
	Offshore	Domestic	Offshore	Domestic
$Electronic\ networks_f$	-0.005***	-0.005***		
	(0.001)	(0.001)		
$\times ln(CAD_j)$	-0.000	0.000**		
	(0.001)	(0.000)		
$\times ln(Servers_c)$	0.002***	0.002***	0.003***	0.003***
	(0.001)	(0.000)	(0.001)	(0.001)
$ln(CAD_j)$	-0.008***	-0.004***		
	(0.002)	(0.001)		
$\times ln(Servers_c)$	0.003***	0.001***	0.004***	0.002***
	(0.001)	(0.000)	(0.001)	(0.000)
$ln(VA\ Prod_f)$	0.002***	0.001***		
	(0.000)	(0.000)		
$Distance^a_{fs}$	-0.026**	-0.019**	-0.018	-0.026
	(0.011)	(0.009)	(0.030)	(0.023)
$ln(Diff\ inputs_j)$	0.007***	0.002***		
	(0.001)	(0.000)		
$\ln(w_s/w_{US})$			-0.021	-0.004
			(0.018)	(0.005)
$ln(Human\ Capital_c)$			-0.051	-0.024
			(0.035)	(0.017)
$ln(Servers_c)$			0.025***	0.009***
			(0.007)	(0.002)
Country fixed effects	yes	yes	no	no
Firm fixed effects	no	no	yes	yes
Adj. R-Squared	0.21	0.11	0.10	0.09
N	270,000	363,900	103,000	1,382,000

Notes: Standard errors clustered by country. There are 185 country clusters in columns 1 and 2, and 70 country clusters in columns 3 and 4. *, **, *** denote 10%, 5% and 1% significance respectively. *a* Distance between a country's main city and the closest manufacturing establishment in the firm. N rounded for disclosure avoidance.

Table 8: First stage regressions for the probability of fragmentation and offshoring

Each column is the first stage regression for the listed endogenous variable

Probability of Fragmentation Regressions							
	Electronic networks only			With CAD interaction			
Instruments	$ln(VAP_i$ Terciles) Q2	Q3	$Elec.Nets_i$ Indicator	$ln(VAP_i$ Terciles) Q2	Q3	$Elec.Nets_i$ Indicator	$\times ln(CAD_j)$
$ln(VAP_i)$ Q2	0.156***	0.089***	0.043***	0.156***	0.089***	0.043***	-0.060***
	(0.006)	(0.004)	(0.005)	(0.006)	(0.004)	(0.005)	(0.007)
Q3	-0.049***	0.391***	0.063***	-0.049***	0.391***	0.063***	-0.082***
	(0.006)	(0.007)	(0.005)	(0.006)	(0.007)	(0.005)	(0.008)
$Elec\ netwks_i$	-0.007*	0.039***	0.263***	-0.012**	0.042***	0.277***	-0.088
	(0.004)	(0.004)	(0.004)	(0.006)	(0.006)	(0.006)	(0.064)
$\times ln(CAD_j)$				-0.004	0.003	0.013***	0.171***
				(0.003)	(0.003)	(0.003)	(0.056)
Adj. R^2	0.08	0.27	0.14	0.08	0.27	0.14	0.51
Shea's Partial R^2	0.03	0.11	0.04	0.03	0.12	0.04	0.03
F-Statistic[a]	1431	374	1217	1083	281	938	736
Probability of Offshoring Regressions							
$ln(VAP_i)$ Q2	0.140***	0.097***	0.013	0.140***	0.097***	0.013	-0.060***
	(0.011)	(0.007)	(0.008)	(0.011)	(0.007)	(0.008)	(0.007)
Q3	-0.080***	0.398***	0.032***	-0.080***	0.397***	0.032***	-0.082***
	(0.012)	(0.011)	(0.010)	(0.013)	(0.011)	(0.010)	(0.008)
$Elec\ netwks_i$	-0.014*	0.049***	0.240***	-0.030**	0.061***	0.267***	-0.088
	(0.008)	(0.007)	(0.006)	(0.014)	(0.011)	(0.012)	(0.064)
$\times ln(CAD_j)$				-0.016	0.013*	0.027**	0.171***
				(0.010)	(0.008)	(0.012)	(0.056)
Adj. R^2	0.09	0.25	0.12	0.09	0.25	0.12	0.51
Shea's Partial R^2	0.04	0.11	0.04	0.03	0.11	0.05	0.04
F-Statistic[a]	570	176	481	462	137	371	403

Notes: Each column reports the coefficients and standard errors for the excluded instruments in the first stage regression of the respective endogenous variable. Instruments are 2002 lagged values. Standard errors clustered by state. *, **, *** denote 10%, 5% and 1% significance respectively. *a* F-Statistic is for a test of the null hypothesis that the instruments are jointly equal to zero.

Table 9: Variance decomposition of wages, technology, and distance

	Fragmentation			Off|Frag		
	Wages	Technology	Distance	Wages	Technology	Distance
Baseline	0.01	0.28	0.01	0.02	0.24	0.02
W/CAD	0.01	0.36	0.01	0.02	0.47	0.02

Notes: Based on the IV Baseline and W/CAD estimates presented in Table 6.

A Theory Appendix–For Online Publication Only

A.1 Derivations of the productivity thresholds

In equilibrium, final good producer i chooses the sourcing location s that maximizes profits $max_s\{\pi_{is}\}$, where $s \in \{I, D, O\}$. Since fragmentation entails a fixed cost, it will never occur if Equation (2.5) does not hold. In this section, I determine the optimal fragmentation strategy for the subsets of producers in a geographic state for whom: (i) domestic fragmentation maximizes variable profits; (ii) offshoring maximizes variable profits. I first determine producers' optimal share of fragmented tasks, and then identify those producers' profit maximizing decision.

Producers who face costs $c_{kiD} < c_{kiO}\ \forall k$ represent the subset of producers for whom domestic fragmentation maximizes variable profits, N_D. Figure A.1a illustrates this cost scenario. In the figure, C_D, the cost of the composite M defined in Equation (2.8), is simply the area under the bold line. Because domestic fragmentation also entails a fixed cost, Figure A.1b depicts the optimal sourcing strategy for firms with these wage and cost conditions. Fragmentation lowers marginal costs and therefore results in a profit function that is steeper in φ, but the fixed cost to fragment means that, of the producers in the set N_D, only those with productivity above the threshold

$$\tilde{\varphi}_D = \left[\frac{\sigma^{1-\varepsilon}}{(1-\sigma)A}\left(\frac{f_D}{C_D^{1-\varepsilon} - w_h^{1-\varepsilon}}\right)\right]^{\frac{1}{\varepsilon-1}}, \tag{A.1}$$

find it optimal to fragment domestically.

The subset of producers for whom offshore fragmentation maximizes variable profits face costs $c_{kiD} > c_{kiO}\ \forall k$. Figure 1a depicts this situation. C_O, the cost of the composite input M, is the area under the bold line. The cost of M under offshoring is clearly lower than the cost with domestic fragmentation, which is the lower than the cost from integrated production. If the relative fixed costs are small compared to the relative costs of M under domestic versus offshore fragmentation, then optimal profits are similar to those in Figure A.1b, except here only integrated production or offshoring take place. However, if relative fixed costs are large compared to relative savings, or

$$\frac{f_O}{f_D} > \frac{C_O^{1-\varepsilon} - w_h^{1-\varepsilon}}{C_D^{1-\varepsilon} - w_h^{1-\varepsilon}},$$

then integrated production, domestic fragmentation, and offshoring are all possible profit maximizing strategies. Figure 1b depicts this case. Producers with productivity between $\tilde{\varphi}_D$ and $\tilde{\varphi}_O$, fragment domestically, while those with productivity above $\tilde{\varphi}_O$ offshore, where

$$\tilde{\varphi_O} = \left[\frac{\sigma^{1-\varepsilon}}{(1-\sigma)A}\left(\frac{(f_O - f_D)}{C_O^{1-\varepsilon} - C_D^{1-\varepsilon}}\right)\right]^{\frac{1}{\varepsilon-1}}. \tag{A.2}$$

A.2 Comparative Statics

The model provides a framework in which to assess how changes in producer technology, distance to suppliers, and labor cost differences affect the decision to fragment production. This section assesses how these factors affect: (i) whether or not fragmentation is potentially feasible (i.e., the impact on variable profits), and (ii) total profits.

A.2.1 Variation in producer's technology

The model predicts that plants with better communication technology, η, will face lower fragmentation costs. In particular, the cost of the composite input M for a producer fragmenting from location s is decreasing in technology, according to:

$$\frac{\partial C_s}{\partial \eta} = \frac{\partial \bar{k}_s}{\partial \eta}\left[\alpha w_s \tau(\bar{k}_s) - w_h\right] + \alpha w_s \int_0^{\bar{k}_s} \frac{\partial \tau(k)}{\partial \eta}\, dk < 0. \tag{A.3}$$

The term in square brackets in Equation (A.3) is equal to zero from Equation (2.7).[37] The second term represents the inframarginal savings that result from better technology. Holding distance and wage differences constant, an improvement in communication technology decreases fragmentation costs. This decrease means that fragmentation is now potentially viable for a larger set of firms.

Producers for whom fragmentation already maximized variable profits are also more likely to fragment production in response to improvements in their communication technology. The change in fragmentation profits from an improvement in technology η is:

$$\frac{\partial \pi_s}{\partial \eta} = (1-\varepsilon) B [C_s]^{-\varepsilon} \frac{\partial C_s}{\partial \eta}, \tag{A.4}$$

where

$$B \equiv \frac{(1-\sigma)A}{(\sigma\varphi)^{1-\varepsilon}}.$$

Plugging in Equation, (A.3), better technology increases fragmentation profits. Since π_I is unaffected by the change, this implies a lowering of the productivity threshold above which fragmentation is optimal.

[37] This is essentially the envelope condition in that the impact of changes in the share of tasks fragmented on profits is zero to the first order. As is true for all derivatives, this expression holds for small changes in η. Figure A.1a shows that the derivative may not capture the effect of large changes in η on task production costs.

A.2.2 Variation in the home wage

An increase in the producer's home wage, w_h, makes fragmentation relatively more profitable. The change in integrated profits relative to fragmented profits is

$$\frac{\partial \pi_I / \partial w_h}{\partial \pi_s / \partial w_h} = \frac{[w_h]^{-\varepsilon}}{(1-\bar{k_s})[C_s]^{-\varepsilon}}. \tag{A.5}$$

Plugging in the equation for C_s and simplifying shows that the decrease in profits from integrated production is always greater than the decrease from fragmented production whenever

$$(1-\bar{k}) + \frac{1}{\tau(\bar{k})} \int \tau(k) dk > (1-\bar{k})^{(1/\varepsilon)}, \tag{A.6}$$

which is a condition that always holds whenever $\bar{k} > 0$.

A.3 Domestic versus offshore sourcing

Of the firms that fragment production, only those with productivity above $\tilde{\varphi}_O^{\varepsilon-1}$ do so offshore. Since the slope of the offshoring profit function depends upon fragmentation costs, the likelihood of exceeding $\tilde{\varphi}_O^{\varepsilon-1}$ is also decreasing in the distance between a firm and its potential offshore sourcing locations. More formally

$$\frac{\partial \tilde{\varphi}^{\varepsilon-1}}{\partial \delta} = \left[\frac{\partial C_O}{\partial \delta} - \frac{\partial C_D}{\partial \delta} \right] \left(\frac{w_h (f_O - f_D)}{\left[C_O^{1-\varepsilon} - C_D^{1-\varepsilon} \right]^2} \left(\frac{\sigma^{2-\varepsilon} A}{(1-\sigma)^2} \right) \left(C_O^{-\varepsilon} - C_D^{-\varepsilon} \right) \right). \tag{A.7}$$

The three terms inside the parentheses are positive, so the effect on the offshoring threshold depends upon the sign of the terms in the square brackets. If a decrease in distance to foreign suppliers does not affect plants' distance to domestic suppliers, then the second term is zero and Equation (A.7) is positive. The offshoring threshold is therefore higher, leading to the following prediction:

The offshoring threshold also depends upon communication technology. Specifically, the effect of changes in technology on the productivity threshold is given by

$$\frac{\partial \tilde{\varphi}^{\varepsilon-1}}{\partial \eta} = \left[\frac{\partial C_O}{\partial \eta} - \frac{\partial C_D}{\partial \eta} \right] \left(\frac{w_h (f_O - f_D)}{\left[C_O^{1-\varepsilon} - C_D^{1-\varepsilon} \right]^2} \left(\frac{\sigma^{2-\varepsilon} A}{(1-\sigma)^2} \right) \left(C_O^{-\varepsilon} - C_D^{-\varepsilon} \right) \right) \tag{A.8}$$

The terms in parentheses are positive, so the offshoring threshold is decreasing in technology as long as $\frac{\partial C_O}{\partial \eta} < \frac{\partial C_D}{\partial \eta}$. Plugging in Equation (A.3) shows that an improvement in communication technology will make offshoring relatively more profitable than domestic fragmentation if the inframarginal cost savings from offshored production exceed the inframarginal cost savings of domestic fragmentation. Consider

the case depicted in Figure 1a where $c_{iD} > c_{iO}$. In this case, offshoring maximizes variable profits, but the higher fixed cost to offshore induces domestic fragmentation over some range of φ. Under these conditions, the terms in the first set of brackets in Equation A.8 can be expressed as

$$\alpha w_O \int_{\bar{k_D}}^{\bar{k_O}} \frac{\partial \tau_O(k)}{\partial \eta}\, dk + \alpha w_O \int_{0}^{\bar{k_D}} \frac{\partial \tau_O(k)}{\partial \eta}\, dk - \alpha w_D \int_{0}^{\bar{k_D}} \frac{\partial \tau_D(k)}{\partial \eta}\, dk. \tag{A.9}$$

The first term is always negative, while the second two terms offset each other if the technology shock affects all tasks and domestic and offshore costs equally. When this occurs, a technology improvement will lower a firm's offshoring threshold, making it more likely that the firm offshores. In contrast, if the technology shock lowers domestic fragmentation costs relatively more than offshoring costs, the offshoring threshold may rise, thereby decreasing the likelihood that a given firm will exceed the threshold.

B Data Appendix–For Online Publication Only

B.1 Contract Manufacturing Question

The CMS purchase question is the third part of the special inquiries section, question 26 on the Census of Manufacturers. The exact question is:

3. Did this establishment purchase contract manufacturing services from other companies or other establishments of your company to process materials or components that this establishment owns or controls?

0496 ☐ Yes, primarily with establishments WITHIN the 50 States and the District of Columbia

0497 ☐ Yes, primarily with establishments OUTSIDE of the 50 States and the District of Columbia

0498 ☐ No

I cannot provide any actual examples of firms or contract manufacturing service (CMS) purchases in the Census data because the data are confidential and respondents' identities cannot be revealed. However, hypothetical examples of CMS purchases include the manufacturing of company A's MP-3 player components as instructed by company A; the assembly of company B's computer processing chips in B's overseas plants using specified inputs and a precise design criterion; and the production of company C's shoes in non-affiliated factories using soles provided and materials specified by company C. In each case, the purchaser furnishes production specifications to the manufacturing service provider.

B.2 Description of the CMS sample

The CM is conducted in years that end in 2 and 7. It covers the universe of manufacturing establishments in the U.S, though not every establishment is asked the CMS question. While all large plants and all plants that belong to multi-unit firms were asked the CMS question, only a random sample of small and medium-sized plants was asked the question. Data for the smallest manufacturing plants, generally those with less than five employees, are based only on administrative records and therefore do not include any information about those plants' CMS purchases.[38] Since the administrative records are often based on imputed data, have no information about CMS purchases, and account for only 1.5% of sales and three percent of employment in the manufacturing sector, I exclude them from the entire analysis. All establishments that receive a census form in the mail are legally required to return the completed form. Despite the legal requirement, a fraction of establishments did not respond to the CMS question.

I assess the observable characteristics of plants outside the CMS sample to address potential issues from sample selection. While the CMS sample covers 54 percent of manufacturing establishments (excluding the administrative records), it includes 75 percent of sales and 71 percent of employment. Table B.1 presents information on plants in the CMS sample relative to those outside the sample. The first three columns provide participation shares and show that while the CMS sample comprises of 54 percent of U.S. manufacturing establishments (excluding administrative records), it covers 75 percent of sales and 71 percent of employment. The last three columns of Table B.1 present summary statistics by CMS non-response type. As expected, establishments not asked the CMS question are substantially smaller than plants in the sample. To address potential selection bias in the analyses, I use a number of establishment-level variables, including sales, employment, age, firm age, multi-unit firm status, industry, and industry employment interactions, to estimate, via logit, the probability that an establishment will be in the CMS sample. I then predict the probability that an observation is in the sample and, following Cameron and Trivedi (2005), use the inverse probability as a weight in the empirical analyses.[39]

[38] The CM uses both short and long form questionnaires, and only the long forms ask the CMS purchase questions. While all large and multi-unit firm establishments receive the long form, only a random sample of small, single-unit firms receive the long form. Data for the smallest establishments is imputed from Federal tax returns and industry averages.

[39] Cameron and Trivedi recommend using weights ("that are inversely proportional to the probability of inclusion in the sample") for a descriptive or data summary approach. They note that if a regression model is correctly specified then sample weighted and unweighted estimates should have the same probability limit, Cameron and Trivedi (2005) pp. 817-21.

B.3 Within industry participation shares

Table B.2 provides the mean and standard deviation of plant participation shares within the 472 six-digit NAICS industries. The industry average of plants that fragment domestically matches the aggregate national share of 0.27, while the average share of plants that offshore within an industry is higher at 0.03. Table B.2 also shows that plant participation shares vary substantially across industries. The standard deviation of the industry share of domestic fragmenters is 0.13, and the standard deviation of the industry offshoring share is 0.04. The second and third panels of the Table depict similar patterns for the average industry shares of sales and employment respectively. Since some of the differences in these participation shares may be attributable to the availability of standardized inputs, the regression analyses include industry fixed effects. When performing a firm-level analysis in which industry fixed effects are not possible due to multi-industry firms, I control for this possibility using a the Nunn (2007) measure of an industry's share of differentiated input use. This share captures the extent to which industry inputs are not market or reference priced.

B.4 Import data

Table B.4 presents information to assess the importance of this customized offshoring. Column 1 indicates that firms with one or more plants that purchase CMS account for 67 percent of imports. Domestic fragmenters import 36 percent of imports, while offshoring firms import 31 percent. Column 2 shows that average firm imports are also significantly higher for offshoring firms. While it is impossible to measure the exact extent to which imports correspond to CMS purchases, firms that purchase CMS clearly dominate import activity. Column 2 in Table B.4 shows that about 40 percent of firms that do not purchase CMS import manufactured goods. Imports by firms that do not purchase CMS may reflect purchases of standardized materials, inputs, or final goods that are sold in the marketplace; or they may reflect imports that relate to activities in other sectors in which the firm is active. About half of all firms that purchase domestic CMS import goods, while 90 percent of offshoring firms import. Firms that purchase CMS offshore but do not import may be offshoring the final assembly of goods that they sell overseas.[40] Nearly all firms that purchase CMS both domestically and offshore import manufactured goods.

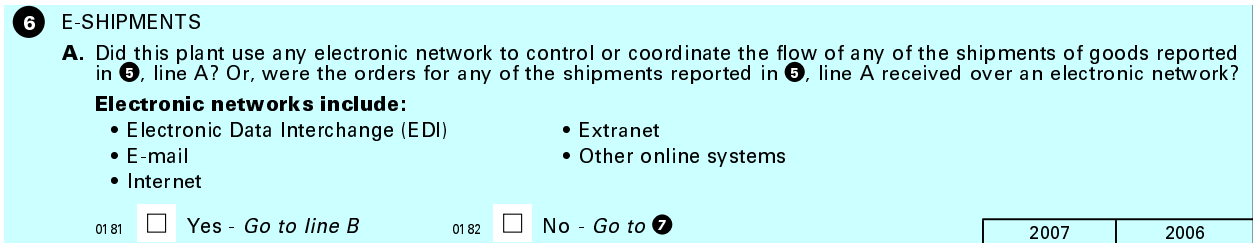
Table B.4 also presents the average extent to which firms offshore, measured as firms' imports over sales. Column 4 shows that domestic fragmenters source a relatively small share of their production offshore. Their average imports over sales is only

[40] Almost all of these firms export goods. It is also possible that some firms are erroneously classified as non-importers due to an inability to link the import data to the census data. This should not be a big issue, however, as I match 92 percent of the transactions and 91 percent of the value of imports.

three percent, compared to 20 percent for firms that primarily offshore. Somewhat surprisingly, firms with no CMS purchases import an average of nine percent of their sales. To assess whether this high share may result from industry compositional differences or sales in other sectors, I calculate firms' share of imports over sales relative to the average share of their modal industry. Excluding firms with employment outside of manufacturing, the relative shares are 0.67, 0.68 and 3.9 for non-purchasers, domestic fragmenters, and offshorers respectively. Offshoring firms' share of imports over sales is almost four times their industry average, while non-purchasers and domestic fragmenters' share is less than their industry mean. Column 5 shows firms' share of imports from low-income countries. I classify countries as low income if they are in the bottom two per-capita GDP terciles.[41] Column 2 shows that firms with no CMS purchases and domestic CMS purchases import 28 and 19 percent of their manufactured good imports from low-income countries respectively. In contrast, offshorers source almost half of their imports from low-income countries.

B.5 Variable Descriptions

Electronic Networks: I measure whether a plant used electronic networks in 2007 with a dummy variable equal to one for plants that report using an electronic network to control or coordinate shipments. The precise question from the 2007 Census of Manufactures is: I measure plants' use of electronic networks in 2002 using data from


6 E-SHIPMENTS

A. Did this plant use any electronic network to control or coordinate the flow of any of the shipments of goods reported in 5, line A? Or, were the orders for any of the shipments reported in 5, line A received over an electronic network?

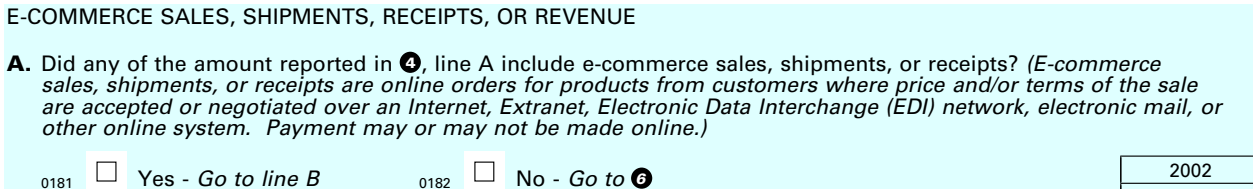
Electronic networks include:

- Electronic Data Interchange (EDI)
- E-mail
- Internet
- Extranet
- Other online systems

0181 ☐ Yes - *Go to line B* 0182 ☐ No - *Go to* 7

2007 | 2006

this similar question in the 2002 Census of Manufactures: Although the question asks


E-COMMERCE SALES, SHIPMENTS, RECEIPTS, OR REVENUE

A. Did any of the amount reported in 4, line A include e-commerce sales, shipments, or receipts? *(E-commerce sales, shipments, or receipts are online orders for products from customers where price and/or terms of the sale are accepted or negotiated over an Internet, Extranet, Electronic Data Interchange (EDI) network, electronic mail, or other online system. Payment may or may not be made online.)*

0181 ☐ Yes - *Go to line B* 0182 ☐ No - *Go to* 6

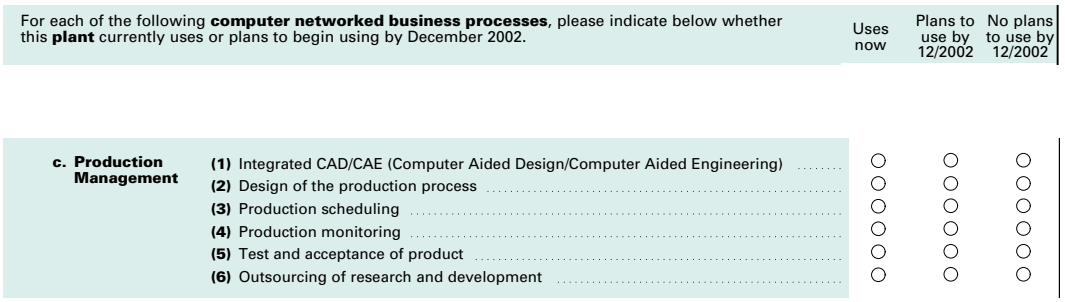
2002

about establishments' use of electronic networks to control or coordinate shipments, data from the the 1999 Annual Survey of Manufactures (ASM) Computer Survey Network Use Supplement (CNUS) show that plants' use of electronic networks to

[41] I obtain countries' per-capita GDP in 2007 from the International Monetary Fund. The GDP data are unavailable for a small number of countries that represent less than one percent of imports in each CMS category.

sell goods is correlated with their use of networks to purchase inputs. I find that plants' acceptance of online orders for their manufactured products has a correlation coefficient of .23 with their use of networks to purchase materials or supplies. In addition, 32 percent of plants that sell goods over networks also use networks to provide information about their design specifications to external suppliers, compared to only 16 percent of plants that do not sell goods over networks. The same pattern (30 percent vs. 16 percent) holds for plants that do or do not use networks to purchase inputs. These findings support the premise that plant use of electronic networks to control or coordinate shipments is a valid proxy for a plant's use of technology to communicate with suppliers.

CAD/CAM Industry Intensity: I measure industry intensity of Computer Aided Design (CAD) and Computer Aided Engineering (CAE) using the Computer Survey Network Use Supplement (CNUS) from the 1999 Annual Survey of Manufactures (ASM). The CNUS asked manufacturing establishments a number of questions about their use of different types of technology. I use the following question to identify whether a particular establishment used CAD/CAE software in 1999: I construct an



For each of the following **computer networked business processes**, please indicate below whether this **plant** currently uses or plans to begin using by December 2002.

		Uses now	Plans to use by 12/2002	No plans to use by 12/2002
c. Production Management	**(1)** Integrated CAD/CAE (Computer Aided Design/Computer Aided Engineering)	○	○	○
	(2) Design of the production process	○	○	○
	(3) Production scheduling	○	○	○
	(4) Production monitoring	○	○	○
	(5) Test and acceptance of product	○	○	○
	(6) Outsourcing of research and development	○	○	○

indicator for all plants that used CAD in 1999 and calculate the share of plants in an industry using CAD relative to plants that did not use CAD and had no plans to use it by 2002. I do not include plants that report planning to use CAD by 12/2002. The ASM is not a representative sample, so I use weights provided in the CNUS to avoid any potential selection bias. I calculate the CAD measure at the NAICS 6 level.

U.S. Wage Data: I use 2006 U.S. wage data from the Bureau of Labor Statistics' Occupational Employment Statistics (OES) Survey to measure production worker wages by state and industry. The state level data provide the mean wage by six-digit occupation. I limit the data to the 110 "Production occupations" and match them to national OES data on occupations and four-digit NAICS industries. I use the national data to determine the occupational intensity of each industry, which I calculate as the share of workers in a given occupation in the industry's total employment of production occupation workers. I match the national share of each occupation within an industry to the state-occupation level wage data. I then compute the state-industry wage as the average, weighted by the national industry share, of each occupation's

wage within an industry. The wage for industry j in state h is then:

$$wage_{hj} = \sum_{o \in j} [wage_{o,h} \times \frac{emp_{o,j,US}}{emp_{j,US}}], \tag{B.1}$$

where o denotes occupations. In principle, this methodology avoids attributing wage differences across industries and states to compositional differences in a state's employees. In practice, occupations are unevenly distributed across states so that some states do not have employment, and therefore wage values, for certain occupations. For example, only three states have nuclear power operators, seven have shoe machine operators, and 15 have semiconductor processors.[42] In addition, some states are missing wage data for some of their occupations. To ensure that the shares of employment in each state-industry combination sum to one, I use the average state wage for the five-digit occupation (or four-digit occupation if the five-digit occupation wage is missing) to which the missing occupation belongs.

Relative Foreign Wages: The foreign wage data are from the International Labor Organization and available for 1983-2003. I use monthly wages in US $s provided by Oostendorp (2005). I construct relative foreign wages for each industry-occupation as w^*/w_{US}, where w^* denotes the wage in a foreign country. This relative wage follows the theoretical framework and provides a unit free measure that applies to specific industries and occupations. To the extent that skill varies by industry and occupation, the relative wage controls for compositional differences in countries' wages that are driven by workers of varying skill levels. I primarily use wages from 2000 because it is the most recent year of complete data. When the 2000 data are missing for a given country-industry-occupation, I use data from the closest year. Because the relative wage is unit free, it is not affected by dollar inflation, though significant changes in the exchange rate over time may cause measurement problems for data substitution from other years. I match the ILO data to NAICS industries by hand and average over industries to obtain a relative foreign wage in a given NAICS industry. The industries vary from three-digit to six-digit NAICS. Despite substituting missing values with data from alternate years, there are still missing data. When available, I replace missing data with the average wage for a higher level of NAICS aggregation in a given country. There are some countries for which no data are available. Since identifying sourcing locations is a primary focus of this paper, country-specific characteristics are critical and I therefore do not impute data for the missing countries. When I construct the firm import-weighted average relative wage, I drop firms for which 50 percent or more of its imports have no wage. For the remaining firms, I re-normalize import shares to construct an import-weighted average relative wage according to:

$$\left(\frac{w^*}{w_{US}}\right)_f = \frac{\sum_{s,j}[imports_{f,s,j} \times \frac{w_{s,j}}{w_{us,j}}]}{imports_f} \tag{B.2}$$

[42]The uneven distribution of occupations across the country is consistent with production fragmentation. It also suggests an important role for the substitutability of labor in a given location, for a given industry. I do not address this dimension of variation as a determinant of fragmentation.

Skill measures: I measure state skill differences using 2007 data from the American Community Survey (ACS). For both total workers in a state and for the subsample of production workers in a state, I calculate a)the share of workers with at least a college degree; b) the share of workers with at least an associate's degree; and c) the share of workers with at least a high school degree.

Deep Water Ports: I identify all potentially relevant water ports using data from the Maritime Administration's Port Import Export Reporting Service (PIERS).[43] The data are collected from vessel manifests and bills of lading and provide imports, measured by number of shipping containers in twenty-foot equivalent units (TEUs), by port. To ensure the ports correspond to viable import channels, I restrict the ports to deep water ports with a value of imports greater than 100 TEUs. (I except Anchorage, Alaska from this exclusion criterion since it imported 92 TEUs in 2007, but is the largest viable import port in Alaska.) Latitude and longitude for each port are from the Intermodal Terminal Facility database from the Research and Innovative Technology Administration's Bureau of Transportation Statistics (RITA/BTS) National Transportation Atlas Databases (NTAD) 2010.[44] This database is missing Port Manatee, FL and the Port of Honolulu, HI so I obtain latitude and longitude for these ports from: `www.worldportsource.com/states.php`. The final port data are available at `http://mba.tuck.dartmouth.edu/pages/faculty/teresa.fort/research.html`.

Border Crossings: I identify all potential border crossings with Canada and Mexico using border crossing/entry data from RITA/BTS.[45] The data originate from the U.S. Department of Homeland Security, Customs and Border Protection, OMR database. There are 86 Canadian and 25 Mexican entry ports into the U.S. I exclude the crossings that had no truck traffic in 2007 to obtain 82 Canadian and 22 Mexican potential crossing points. I attach latitudes and longitudes to these crossings using the centroid for the county in which the port is located. The final data are available at `http://mba.tuck.dartmouth.edu/pages/faculty/teresa.fort/research.html`.

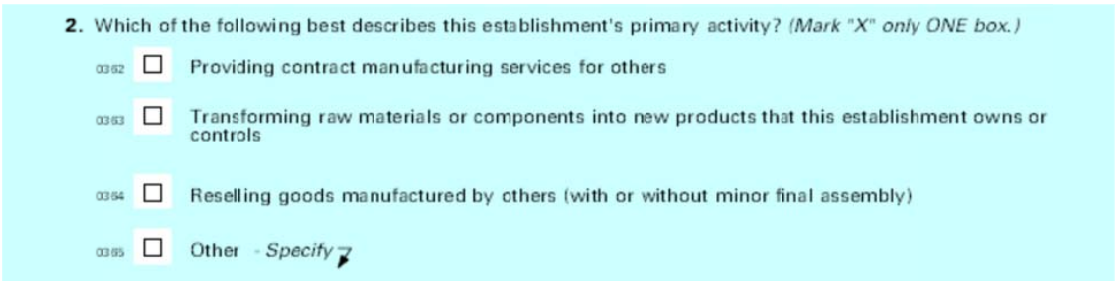
Domestic suppliers The CMS sample also has information about plants' primary activity. Treating all plants that identify their primary activity as "Providing contract manufacturing services to others", I calculate the distance between each manufacturing plant in the CMS sample and the closest manufacturing service provider (MSP). The precise question I use to identify MSPs is provided below.

I also calculate an alternative domestic distance measure as the weighted average of the minimum distance between a plant and all of its input suppliers. I use the 2002 BEA I-O tables to identify each plant's inputs based on the plant's four-digit

[43]The data are available here: `www.marad.dot.gov/library_landing_page/data_and_statistics/Data_and_Statistics.htm`.

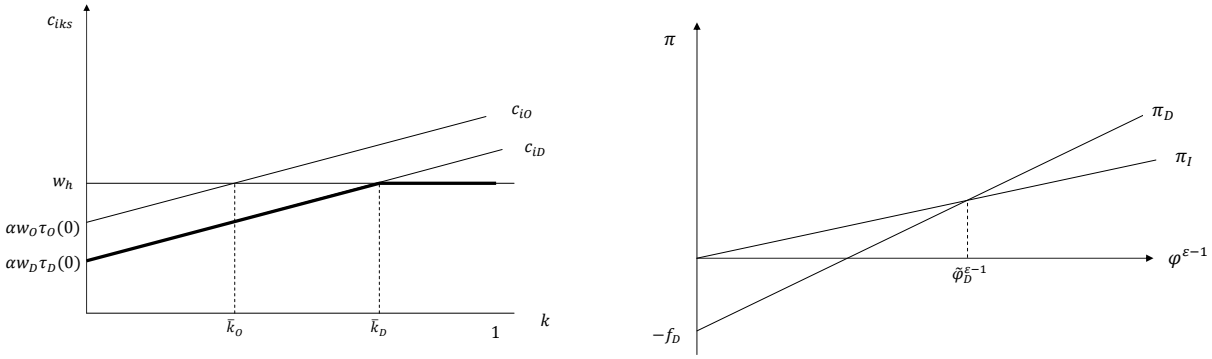
[44]Data available here: `www.bts.gov/publications/national_transportation_atlas_database/`.

[45]The data can be downloaded here: `http://www.bts.gov/programs/international/transborder/TBDR_BC/TBDR_BCQ.html`.



NAICS industry. I also use the I-O tables to calculate the share of each input in production, which I use as a weight. The weighted distance to input suppliers is positively correlated with the distance to the closest MSP. Using the weighted distance in regression analyses in lieu of the MSP distance measure yields similar results.

Figure A.1: Integrated production and domestic fragmentation



(a) Variable costs per task (b) Profits

Table B.1: Plant characteristics by response status

	Participation Shares			Means		
	Plants	Sales	Emp	Sales[a]	Emp	ln(VAP)
In CMS Sample	0.54	0.75	0.71	36,778	86	4.56
Out of CMS Sample						
Not Answered	0.21	0.23	0.24	29,548	77	4.61
Not Asked	0.25	0.02	0.05	2,314	13	4.25
No Info	0.00	0.00	0.00	21,147	61	4.75
All Manufactures	1.00	1.00	1.00	26,638	66	4.50

Table B.2: Summary statistics for industry participation shares by CMS purchase status

	Plants		Sales[a]		Emp[a]	
	mean	sd	mean	sd	mean	sd
No Purchases	0.70	0.14	0.69	0.14	0.63	0.18
Domestic Purchases	0.27	0.13	0.28	0.13	0.32	0.17
Offshore Purchases	0.03	0.04	0.04	0.05	0.05	0.09

Table B.3: Plant participation shares within average state wage terciles, by CMS purchase status

	State wage tercile		
	Low	Medium	High
No Purchases	0.76	0.73	0.68
Domestic Purchases	0.22	0.25	0.30
Offshore Purchases	0.02	0.02	0.02

Wage terciles based on average production worker wages within a state.

Table B.4: Manufacturing firm imports, by contract manufacturing service purchase status

	Share of total	Imports[a] (mean)	Importer Share	$\frac{Imports}{Sales}$ (mean)	Low income (share)
No Purchases	0.33	2.47	0.41	0.09	0.28
Domestic Purchases	0.36	7.16	0.53	0.03	0.19
Offshore Purchases	0.07	18.74	0.90	0.20	0.48
Domestic & Offshore	0.24	428.53	$\approx 1^b$	0.16	0.19

Share of total decomposes total imports by firms in the CMS sample by category. Importer share is the fraction of importing firms. $\frac{Imports}{Sales}$ is the average of total firm imports over sales. Low income is firms' average share of low income imports. *a* Imports in millions of $s. *b* Rounded for disclosure avoidance.

Table B.5: Distance to potential suppliers

	Domestic[a] Suppliers	Deep water Ports	Border Crossings Canada	Mexico
No Purchases	2.3	286	469	992
Domestic Purchases	2.2	302	431	1,028
Offshore Purchases	2.0	248	483	950
All Plants	2.2	290	459	1,001

a Weighted distance to plant's input suppliers. Inputs and weights based on plant's industry and each input's relative importance in IO tables.

Table B.6: Plant use of electronic networks, by year and 2007 CMS purchase status

	Share of Plants 2002	2007
No Purchases	0.17	0.36
Domestic Purchases	0.22	0.51
Offshore Purchases	0.33	0.61

Share of plants that use electronic networks to control or coordinate shipments by contract manufacturing services purchase status.

Table B.7: Correlations between industry CAD intensity and industry-level variables

	CAD	ShrDesSup	ShrDesCo	KI	SI	DI	Off
Share designs w/suppliers	0.49 *0.00*						
Share designs w/comp. units	0.54 *0.00*	0.23 *0.00*					
Capital Intensity	0.09 *0.04*	-0.24 *0.00*	0.20 *0.00*				
Skill Intensity	0.16 *0.00*	0.25 *0.00*	0.10 *0.02*	0.00 *0.99*			
Diff Inputs	0.26 *0.00*	0.35 *0.00*	0.12 *0.01*	-0.37 *0.00*	0.37 *0.00*		
Wholesale imports to sales	-0.03 *0.57*	-0.02 *0.62*	0.05 *0.27*	-0.08 *0.10*	0.01 *0.92*	0.15 *0.00*	
Whole/Manf imports to sales	-0.03 *0.54*	-0.02 *0.70*	0.10 *0.03*	-0.07 *0.13*	-0.01 *0.81*	0.16 *0.00*	0.92 *0.00*

Notes: Correlation coefficients between industry CAD intensity, the share of manufacturing plants that use electronic networks to share designs with suppliers, the share that use networks to share designs with other company units, capital intensity, skill intensity, share of differentiated inputs, and wholesale firm imports relative to domestic manufactures' sales. Details on industry variables are provided in the text. P-values reported in italics.

Table B.8: Probability that a plant fragments production

Dependent variable is 1 if plant i in industry j and state h fragments production

	Demand	Skill	10+ years	Adopters	Manuf Conc
$Electronic\ networks_i$	0.255***	0.256***	0.254***	0.168***	0.255***
	(0.038)	(0.037)	(0.043)	(0.052)	(0.037)
$\times ln(CAD_j)$	0.066***	0.066***	0.063**	0.034	0.067***
	(0.025)	(0.025)	(0.028)	(0.028)	(0.025)
$ln(wage_h)$	0.167***	0.146***	0.158***	0.119***	0.155***
	(0.030)	(0.033)	(0.029)	(0.038)	(0.033)
$ln(VAProd_i)$ Q2	0.121***	0.120***	0.116***	0.074*	0.121***
	(0.023)	(0.023)	(0.025)	(0.039)	(0.023)
Q3	0.159***	0.157***	0.159***	0.060***	0.158***
	(0.013)	(0.013)	(0.015)	(0.021)	(0.013)
MSP is 5-20 miles away	-0.023***	-0.021***	-0.023***	-0.020***	-0.021***
	(0.005)	(0.004)	(0.004)	(0.006)	(0.004)
20+ miles away	-0.032***	-0.028***	-0.028***	-0.020*	-0.028***
	(0.006)	(0.006)	(0.005)	(0.011)	(0.006)
Port is 51-200 miles	-0.009**	-0.004	-0.003	0.009	-0.007
	(0.004)	(0.004)	(0.005)	(0.008)	(0.004)
200+ miles away	0.004	0.012**	0.013**	0.017**	0.007
	(0.005)	(0.006)	(0.005)	(0.007)	(0.005)
50+ miles to border with					
Mexico	-0.032***	-0.033***	-0.033***	-0.096***	-0.034***
	(0.005)	(0.005)	(0.012)	(0.011)	(0.005)
Canada	-0.006	-0.007	-0.003	-0.007	-0.007
	(0.007)	(0.008)	(0.008)	(0.009)	(0.009)
$ln(BEAIncome)$	-0.003**				
	(0.002)				
$ln(Share\ college_h)$		0.022			
		(0.018)			
$ln(Share\ manuf_h)$					0.005
					(0.009)
NAICS 6 Controls	yes	yes	yes	yes	yes
Adj .R-Squared	0.08	0.08	0.08	0.07	0.08
N	71,600	71,600	62,800	22,200	71,600

Notes: IV regressions that instrument for electronic networks and productivity using lagged 2002 values. Standard errors clustered by state. *, **, *** denote 10%, 5% and 1% significance respectively. Demand controls for personal income in the plant's BEA Economic Area. Skill controls for the share of workers with a college degree. 10+ years is the sub-sample of plants that have existed in the same physical location for at least 10 years. Adopters is the subset of single unit firms that did not fragment in 2002. Manuf conc controls for the share of the population employed in manufacturing. CAD is the CAD/CAM intensity in a plant's industry. MSP denotes manufacturing service provider. Results are also robust to clustering by industry.

Table B.9: Probability that a plant offshores its fragmented production

Dependent variable is 1 if fragmenting plant i in industry j and state h offshores

	Demand	Skill	10+ years	Adopters	Manuf Conc
$Electronic\ networks_i$	0.056	0.055	0.061*	0.027	0.056
	(0.035)	(0.035)	(0.037)	(0.082)	(0.035)
$\times ln(CAD_j)$	-0.051	-0.052	-0.039	-0.147*	-0.051
	(0.038)	(0.038)	(0.035)	(0.086)	(0.038)
$ln(wage_h)$	-0.125***	-0.144***	-0.115**	-0.156*	-0.091**
	(0.045)	(0.046)	(0.047)	(0.080)	(0.044)
$ln(VAProd_i)$ Q2	-0.004	-0.004	0.005	-0.031	-0.004
	(0.019)	(0.020)	(0.019)	(0.057)	(0.020)
Q3	0.035***	0.035***	0.042***	0.050*	0.035***
	(0.011)	(0.011)	(0.012)	(0.028)	(0.011)
MSP is 5-20 miles away	0.002	0.001	0.001	-0.009	0.000
	(0.005)	(0.005)	(0.005)	(0.009)	(0.005)
20+ miles away	0.004	0.002	0.008	0.010	0.001
	(0.006)	(0.006)	(0.006)	(0.013)	(0.006)
Port is 51-200 miles	-0.010	-0.010	-0.014*	-0.016	-0.009
	(0.008)	(0.008)	(0.008)	(0.012)	(0.008)
200+ miles away	-0.021***	-0.019***	-0.023***	-0.027***	-0.016***
	(0.005)	(0.005)	(0.005)	(0.008)	(0.006)
50+ miles to border with					
Mexico	-0.097***	-0.095***	-0.084***	-0.083***	-0.094***
	(0.017)	(0.017)	(0.015)	(0.026)	(0.018)
Canada	-0.010	-0.010*	-0.010	-0.009	-0.005
	(0.007)	(0.006)	(0.007)	(0.014)	(0.006)
$ln(BEAIncome)$	0.002				
	(0.002)				
$ln(Share\ college_h)$		0.033**			
		(0.016)			
$ln(Share\ manuf_h)$					-0.020***
					(0.006)
NAICS 6 Controls	yes	yes	yes	yes	yes
Adj .R-Squared	0.07	0.07	0.07	-0.03	0.07
N	21,500	21,500	19,100	5,300	21,500

Notes: IV regressions that instrument for electronic networks and productivity using lagged 2002 values. Standard errors clustered by state. *, **, *** denote 10%, 5% and 1% significance respectively. Demand controls for personal income in the plant's BEA Economic Area. Skill controls for the share of workers with a college degree. 10+ years is the sub-sample of plants that have existed in the same physical location for at least 10 years. Adopters is the subset of single unit firms that did not fragment in 2002. Manuf conc controls for the share of the population employed in manufacturing. CAD is the CAD/CAM intensity in a plant's industry. MSP denotes manufacturing service provider. Results are also robust to clustering by industry.